

Non-Invasive Diagnosis of Pulmonary Hypertension: The Role of ECG and SCG in Assessing S2 Deviations

Garner, Natasja Toftgaard¹ ; Jensen, Frederik Fruergaard¹ ; Post, Maibritt¹ ; Rytter, Elisa Faaborg¹

¹ Hst-24-st-7-7401, Biomedical Engineering and Informatics MSc,

Department of Health Science and Technology, Aalborg University, Aalborg, Denmark

Abstract—Pulmonary hypertension (PH) is a condition in which pressure increases in the pulmonary arteries. PH can lead to inconsistency in the closure of the pulmonary valve, caused by eccentric hypertrophy and elevated pulmonary pressure. The aim of this study was to investigate if deviations in the second heart sound (S2) in seismocardiography (SCG) signal and end of the T-wave (Tend) in electrocardiogram (ECG) signal can predict PH.

Methods were developed to detect Tend in the ECG signal, and start of S2 (S2onset) in the SCG signal. These detections were used to investigate deviations in Tend and S2onset. The intervals between R-peak and Tend were investigated to determine the validity of the data. The data was gained from an animal model, where pigs were induced with pulmonary hypertension by exposing the subjects to hypoxemia and hypercapnia. Each analysis compared a baseline recording and its corresponding intervention for each subject.

The tests were performed on 9 pigs, where 17 recordings of hypoxemia and 16 recordings of hypercapnia were included. Results from S2onset-Tend shows a mean difference of -32 ms ($p < 0.001$) for hypercapnia and a mean difference of -16 ms ($p = 0.09$) for hypoxemia, making the results from hypercapnia statistically significant.

A significant deviation of S2 compared to Tend, for pigs induced with hypercapnia, is found. Results from hypoxemia shows a tendency, but no statistical significant deviation. Before using SCG and ECG in combination as methods to diagnose PH, further research is required.

I. INTRODUCTION

Pulmonary hypertension (PH) is characterized by increased resistance in the pulmonary vasculature, leading to elevated pulmonary arterial pressure [1], [2]. The increased pressure can affect the functionality of the pulmonic valve and the structure of the right ventricle (RV) [3, p. 123] [4]. World Health Organization has classified PH into five subgroups based on the underlying pathogenesis [5].

- 1) Pulmonary arterial hypertension (PAH)
- 2) Pulmonary hypertension due to left heart disease
- 3) Pulmonary hypertension due to lung diseases and/or hypoxia
- 4) Chronic thromboembolic pulmonary hypertension (CTEPH)
- 5) Pulmonary hypertension with unclear multifactorial mechanisms

In individuals with PH, due to lung diseases, the pulmonary vasculature is affected by vasoconstriction, increasing pulmonary vascular resistance [6], [7]. To adapt to the increased pressure in the pulmonary vasculature, RV performs an adaptive remodeling by concentric hypertrophy [8]. Concentric hypertrophy increases the thickness of the RV wall, increasing the pressure of the RV [8]. When the RV wall enlarges, it leads to a delayed electrical activation of the RV contraction [8]. This will lead to delayed closure of the pulmonic valve (P2) [3, p. 123]. When PH deteriorates, adaptive remodeling of the RV becomes insufficient leading to maladaptive remodeling [8]. This leads to eccentric hypertrophy that increases the volume of blood in the RV [8]. Over time, the heart cannot continue to maintain a higher pressure in the RV leading to premature P2 [9].

In 2024, the procedure for diagnosing PH includes electrocardiograms (ECG), echocardiograms, and right heart catheterization [10]. The invasive method of right heart catheterization can provide

information on pulmonary arterial pressure (PAP) by catheterizing with a balloon flotation pulmonary artery catheter into the right heart side [11]. If the PAP rises above 25 mmHg a patient can be diagnosed with PH [11]. Non-invasive methods such as ECG and echocardiograms can provide information on changes in electrical activities in the heart or blood flow through the heart [10]. In the interpretation of the echocardiogram, healthcare professionals utilizes subjective methods based on the imaging of the blood flow to determine PH [12]. The ECG alone can not diagnose a patient with PH, since changes in the ECG often occur later in the course of the disease [13]. However, it can be used as an additional method to evaluate the severity of the heart's condition, providing information on the progression of the disease [13], [14].

There continues to be a need for an objective and non-invasive method to accurately diagnose PH, to limit hospitalization and misdiagnosis. A study suggested that the start of the second heart sound (S2onset) can be used to detect the end of the T-wave (Tend) using phonocardiogram and ECG, since a correlation under normal conditions was found [15]. Because of this correlation, it would be valuable to investigate how the relationship, between S2onset and Tend, deviates during PH. Instead of phonocardiogram this study will use seismocardiography (SCG) recordings in comparison with ECG recordings [16]. SCG is a non-invasive method to record the mechanical activity of the heart [16]. The readings of SCGs are caused by the mechanical activity such as valves closing, the contraction of the heart, the ejection of blood, and the rapid filling of the left ventricle, which all causes vibrations [16].

Various methods exist for detecting Tend, including using geometric and angle-based approaches [17], [18]. Similarly, S2onset can be detected by using methods that involve thresholds or signal enhancement such as wavelet transforms and Shannon energy analysis [15], [19], [20]. The aim of this study was to investigate if deviations in S2onset in SCG signals and Tend in ECG signals can predict pulmonary hypertension.

II. METHODS

A. Data acquisition

Data was acquired from experiments conducted on 11 pigs [21]. The datasets were given by the supervisors of this study. The experiments got ethical approval from the Danish Animal Experiments Inspectorate, with the license number: 2021-15-0201-01053 [21]. The pigs were mechanically ventilated whilst being under anesthesia. The systolic blood pressure was recorded with right heart catheterization in the right ventricle, as a substitute for PAP. To induce PH in the pigs, they were exposed to hypercapnia and hypoxemia during the experiments. Hypercapnia was induced by adding carbon dioxide into the inhalation. Hypoxemia was induced by replacing the oxygen with nitrogen in the inhalation. [21]

SCG- and ECG data were recorded simultaneously with a sampling frequency of 5000 Hz. The SCG data were recorded with a triaxial accelerometer placed over the fourth intercostal space next to the sternum, where data from the z-axis was used. During the experiments,

carbon dioxide or nitrogen were removed when the systolic pressure reached a plateau or if the pigs suffered severe arrhythmia. When the pigs reached a post intervention normality in their measurements, the experiments were repeated. [21]

B. Filtering of ECG and SCG

Before preprocessing the ECG and SCG signals, the signals were downsampled from the original 5000 Hz to 500 Hz, which is elaborated in appendix I. The filtering of the ECG was based on a frequency analysis. The first frequency analysis showed a continuous 50 Hz frequency band. A highpass finite impulse response (FIR) filter with an order at 70, and cutoff frequency at 0.5 Hz, was implemented in order to reduce low-frequency noise like respiration. A lowpass FIR filter with an order at 70, was adjusted to a cutoff frequency at 40 Hz, to obtain a satisfied signal where the P-, R- and T-wave were clearly identifiable as seen in the second plot in figure 1.

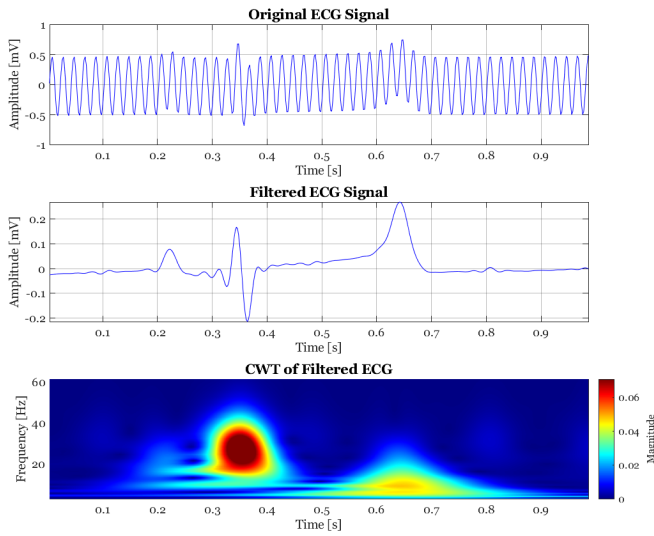


Fig. 1. This figure shows the original and filtered ECG signal, and a scalogram of the filtered ECG.

The SCG signal had a high amount of low frequency activity between the first heart sound (S1) and S2 as seen in the upper plot in figure 2. To make S1 and S2 distinguishable, a continuous wavelet transform (CWT) analysis was performed to produce a scalogram, seen in the second plot in figure 2. Based on the scalogram a FIR highpass filter with hamming window was applied to reduce the leakage of the signal, where the cutoff frequency of the signal was 40 Hz and the filter order was set to 70. This resulted in two clear heart sounds as seen in the third plot in figure 2

C. Detection of ECG attributes

In order to investigate deviations in S2onset compared to Tend, it was necessary to detect the P-, R- and T-wave peaks, and Tend in the ECG signal. To detect the R-peaks, a CWT analysis was performed, to design a filter to isolate the QRS-complexes. The scalogram can be seen in figure 1, where the QRS-complex had a high frequency magnitude around 20-40 Hz. Therefore, a FIR bandpass filter was designed with forward-backward filtering to ensure no phase distortion. The filters cutoff frequencies were chosen from 14 to 40 Hz. The effect of the filter can be seen in figure 3 in the isolated QRS plot. To clarify the R-wave, the QRS signal is rectified in equation 1 and then squared as in equation 2.

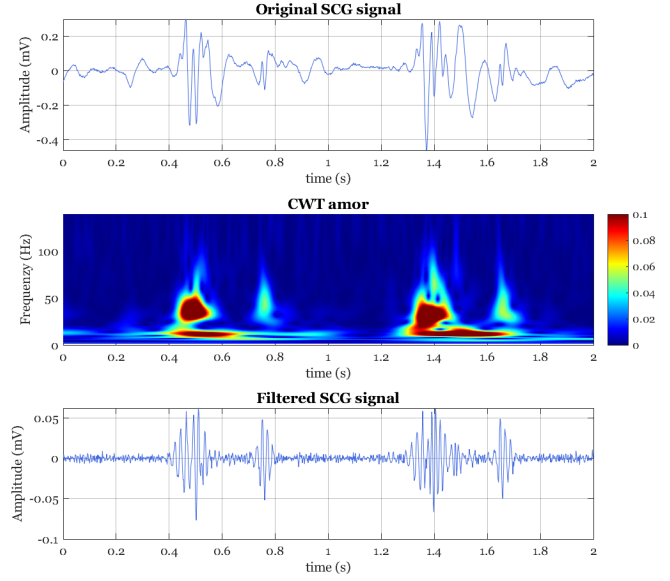


Fig. 2. This figure shows the unfiltered SCG signal with a corresponding scalogram, and the filtered SCG signal.

$$QRS_{nonzero}(t) = \begin{cases} ECG(t), & \text{if } ECG(t) \geq 0 \\ 0, & \text{if } ECG(t) < 0 \end{cases} \quad (1)$$

$$QRS_{squared}(t) = QRS_{nonzero}(t)^2 \quad (2)$$

The R-peaks were detected when a peak in the isolated, rectified and squared QRS signal had a minimum height of 0.01 mV and occurred later than minimum 0.2 seconds after the last detected R-peak. The thresholds were decided based on the rectified and squared signal in the second plot in figure 3.

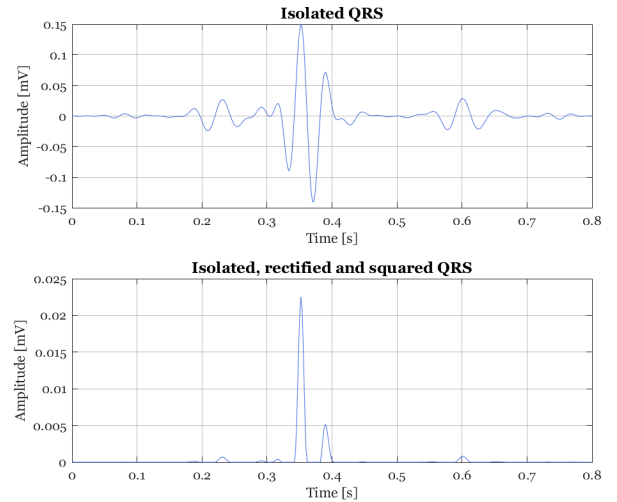


Fig. 3. The isolation of the QRS complex

From the R-peaks the P- and T-peaks were found. The P-peak

was found in the interval 0.7-0.17 s before the R-peak. The T-peak was found in the interval 0.15-0.35 s after the R-peak. This required a heart rate (HR) under 115 bpm. In some cases the subjects exceeded this HR under intervention, leading to a shorter RR-interval. Therefore, if the HR exceeded 115 bpm up to 200 bpm the T-wave was found in the interval 0.1-0.25 s after the R-peak and the P-wave was found in the interval 0.15-0.05 s before the R-peak. There were cases where the subject had an extended plateau phase before the T-wave leading to a delayed T-wave, that required a delayed searching window from 0.25-0.55 s. The functions for finding the ECG events is elaborated further in appendix I

To detect the Tend, the midpoint between the T-peak and the P-peak in the next cardiac cycle was used. From the midpoint a line vector was drawn to the T-peak. The Tend point was defined as the point on the ECG signal under the line vector with the longest difference in distance to the line. This method is illustrated in figure 4. The described method to detect the ECG attribute Tend could also be seen in the flow diagram in figure 5.

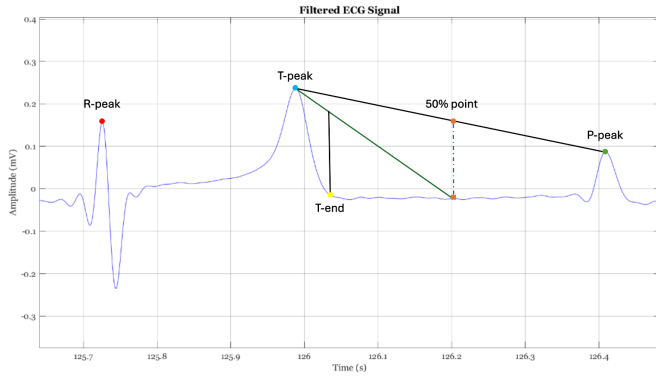


Fig. 4. The method to find the Tend point. The method includes the R-, T- and P-peak, 50% point (midpoint) and the Tend.

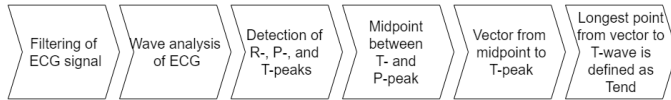


Fig. 5. Flow diagram of ECG data processing

D. Detection of SCG attributes

To detect the attributes of the SCG signal an envelope of the Shannon Energy was made as seen in figure 7, and a function to find the heart sounds was used. This function created an array of binary operators, which marked when an envelope signal of the SCG reached a specific height for a given duration which is specified in appendix I. To distinguish between S1 and S2, the detected R-peak in the ECG was used, since S1 occurs after the R-wave. The midpoint between the R- and T-peak in the ECG was used as a searching point for S2 in the SCG. The steps in the described method can be seen in figure 6.

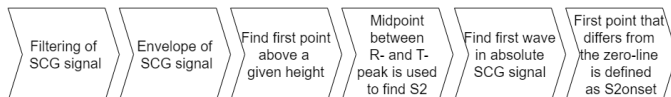


Fig. 6. Flow diagram of SCG data processing

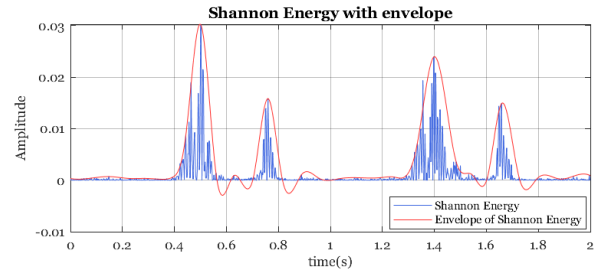


Fig. 7. The Shannon Energy of the SCG signal with an envelope.

E. Definition of time intervals

The detected attributes from the ECG and SCG were used to compare different intervals. The comparison from S2onset to Tend was made to examine deviations from the baseline to the intervention. A corrected R-Tend method was used to predict how the interval changes during different HRs in order to compare the baseline with the intervention. To further understand the deviation in S2onset-Tend, a comparison of the intervals from the R-peak to S2onset and R-peak to Tend was made.

1) **S2onset and Tend:** The detected Tend and S2onset were compared by deducting Tend from S2onset for each cardiac cycle. The method was performed on both a baseline recording and an intervention recording for either the induced hypoxemia or hypercapnia. The mean values for each recording were found and compared.

2) **Corrected R-Tend:** When the HR increases the RR- intervals decreases resulting in a shorter QT-interval [22]. With Fridericia correction formula the QT-interval could be interpreted at different HRs [22]. The QT-interval would originally be calculated from the Q-wave to Tend, however the event of the Q-wave was not detected in the ECG: Therefore, the R-peak was used as a replacement for the Q-wave. The adjusted correction formula used was as described in equation 3.

$$RTc = \frac{RT}{\sqrt[3]{RR}} \quad (3)$$

3) **Comparison of R-Tend and R-S2onset:** In order to understand the S2onset-Tend, a comparison of R-Tend and R-S2onset was made. These intervals presented the change in the relationships during intervention. The comparison was to ensure that the mean calculated from S2onset-Tend was caused by a change in S2onset and not in Tend.

F. Exclusion criteria

To ensure the results gained from the comparisons were accurate, the following exclusion criteria were made:

- | | |
|-------------------------------------|----------------------------|
| 1) Baseline detected beats < 50 | 3) Biphasic T-wave |
| 2) Intervention detected beats < 50 | 4) Late T-wave and high HR |
| | 5) Missing signal |
| | 6) P-wave hidden in T-wave |

'Baseline and intervention detected beats', were the number of beats, where all the attributes were detected. Data which fulfills the exclusion criteria are excluded. Elaborations of the exclusion criteria are in appendix I.

G. Statistics

To determine if the results were significant, a two sided paired t-test was performed for the four different events: S2onset-Tend, R-S2onset, R-Tend, and R-Tc. This indicated whether the result varied significantly from the baseline. The t-test was made using the baseline measurements as a reference method in a paired t-test, and the second method is the intervention measurements. From this a p-value was derived to determine the results to be significant if $p < 0.05$. A confidence interval was presented to indicate how the timing of the intervals had deviated during the intervention.

III. RESULTS

From 71 sessions of either hypercapnia or hypoxemia 33 were included with 16 hypercapnia sessions and 17 hypoxemia sessions. The 38 excluded sessions were excluded based on the following exclusion criteria as seen in table I and II.

TABLE I

INCLUDED AND EXCLUDED DATASETS		
	Included	Excluded
Hypercapnia	16	20
Hypoxemia	17	18
Total	33	38

TABLE II

EXCLUSION CRITERIA

Exclusion criteria	Amount	Percent %
Baseline detected beats <50	4	11 %
Intervention detected beats <50	4	11%
Biphasic T-wave	5	13%
Late T-wave and high HR	4	11%
Missing signal	17	45%
P-wave hidden in T-wave	4	11%

A. R-Tend and R-S2onset results

For R-Tend in table IV the mean of hypercapnia has a significant difference of -34.3 ms from the baseline. The data for hypoxemia is not significantly different from baseline. For R-S2onset in table V the mean of the interventions have a significant difference of -58.4 ms for hypercapnia and -18.2 ms for hypoxemia from the baseline. Figure 8 presents the R-S2onset and R-Tend intervals compared to the systolic pressure over time. Here, the R-Tend is minimally affected by the increase in pressure, whereas R-S2onset decreases.

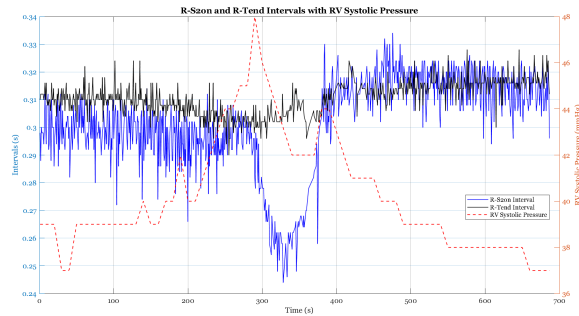


Fig. 8. Systolic pressure, R-Tend and R-S2onset intervals over time during intervention. The plot was made on subject G10 during hypoxemia

Figure 9 presents the boxplot for R-Tend and R-S2onset. The median for R-Tend for both interventions are lower compared to

the baseline, but with a larger standard deviation. The median for R-S2onset for both interventions are lower compared to the baseline.

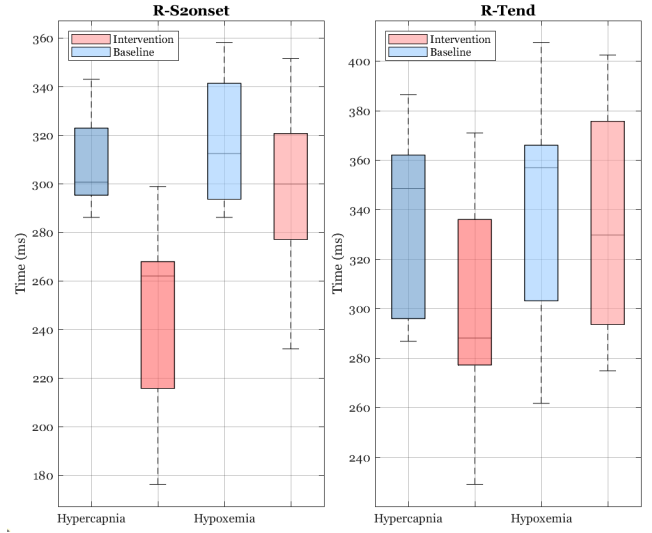


Fig. 9. R-Tend and R-S2onset boxplot of hypercapnia and hypoxemia during baseline and intervention. The blue boxplots represents the baselines and the red represents the intervention.

B. RTc results

For RTc in table VI the mean of the hypercapnia has a difference of -21.5 ms, and the mean of hypoxemia has a difference of -3.8 ms from the baseline. Both interventions are not significantly different from the baseline. In figure 10 the RTc intervals are compared to the systolic pressure over time. The figure shows changes in the RTc interval during the intervention, but throughout the data the changes are inconsistent as seen in the 95% confidential interval.

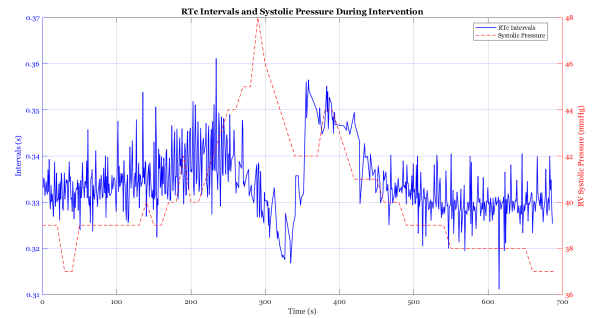


Fig. 10. Systolic pressure and RTc interval over time during intervention. The plot was made on subject G10 during hypoxemia

Figure 11 presents the boxplot for the RTc intervals. The median for RTc for both interventions are lower compared to the baseline.

C. S2onset-Tend results

For S2onset-Tend in table VII the mean of hypercapnia has a significant difference of -32 ms from baseline. The data for hypoxemia are not significantly different from baseline.

Figure 11 presents the boxplot for the S2onset-Tend intervals. The median for hypercapnia is lower during the intervention compared to the baseline, but with a larger standard deviation. The median for hypoxemia is higher during the intervention compared to the baseline.

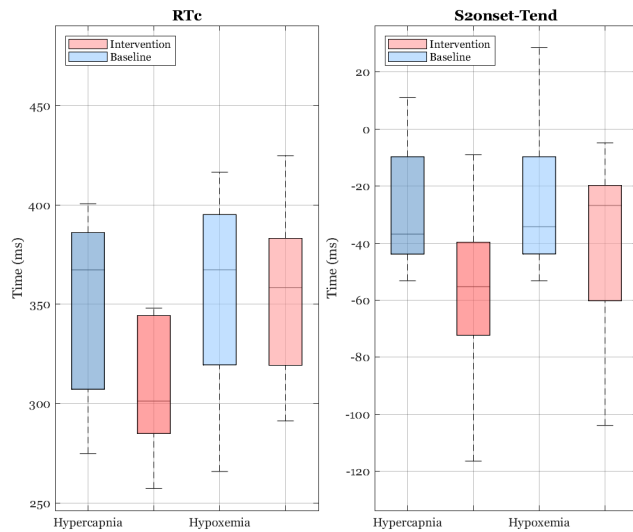


Fig. 11. RTc and S2onset-Tend boxplot of hypercapnia and hypoxemia during baseline and intervention. The blue boxplots represents the baselines and the red represents the intervention.

D. Change in HR and RV systolic pressure

In table III the HR and RV systolic pressure are presented with the means and differences from the interventions compared to the baselines. The HR increases with 27.5 bpm during hypercapnia and 20.7 bpm during hypoxemia. The RV systolic pressure increases with 22.8 mmHg during hypercapnia and 10 mmHg during hypoxemia.

TABLE III
HR AND RV SYSTOLIC PRESSURE MEANS AND DIFFERENCE IN
BASELINE TO HYPOXEMIA AND HYPERCAPNIA

HR				
	Mean (bpm)	Std. Dev. (bpm)	Diff. (bpm)	Diff. (%)
Baseline before hypercapnia	69.2	0.8		
Hypercapnia	96.8	12.6	27.5	39.8%
Baseline before hypoxemia	68.4	0.8		
Hypoxemia	89.1	10.9	20.7	30.3%
RV systolic pressure				
	Mean (mmHg)	Std. Dev. (mmHg)	Diff. (mmHg)	Diff. (%)
Baseline before hypercapnia	30.5	0.4		
Hypercapnia	53.2	4.0	22.8	74.8%
Baseline before hypoxemia	31.4	0.3		
Hypoxemia	41.3	3.7	10.0	31.8%

IV. DISCUSSION

A. Interpretation of the results

1) *R-S2onset and R-Tend interval*: The R-S2onset from table V and R-Tend from table IV, presents the mechanical aspect of S2 and the electrical of Tend. R-S2onset shows that the S2onset occurs significantly earlier for both hypercapnia and hypoxemia recordings, compared to the baseline. This indicates that S2onset occurs earlier during the interventions compared to the baselines. As mentioned in section I eccentric hypertrophy can result in a possible premature closure of P2. Although, the subjects do not

have eccentric hypertrophy, they instead experienced an acute state of ischemia, which increases the PAP [23]. Increased PAP shortens the time needed for P2 to occur, which could cause P2 to occur prematurely [24]. Despite the significant difference from this study, it is unknown whether it is caused by a premature closure of the aortic valve (A2) or P2. A method to investigate the timing of A2 and P2 can give information on which valve is affected. Figure 8 shows that the intervals of R-S2onset are wider than the intervals of R-Tend. This could indicate that there are more detection errors in R-S2onset, where the detection of S2onset occurs earlier or later than the actual S2onset. Even though the intervals for R-S2onset are wide, a general tendency can be observed in relation to how R-S2onset behaves compared to R-Tend. The greater difference of R-S2onset compared to the difference of R-Tend, indicates that the mechanical aspect from S2 is more affected by the systolic pressure or HR than the ECG.

R-Tend shows that Tend occurs significantly earlier for hypercapnia compared to the baseline. As mentioned in section II-E the RR-intervals decreases when the HR increases resulting in a shorter R-Tend interval. As seen in table III the HR increases with 27.5 bpm during hypercapnia, which can explain the decrease in R-Tend. But the HR also increases with 20.7 bpm for hypoxemia, which did not have a significant difference in R-Tend intervals. The table shows a higher difference in RV systolic pressure during hypercapnia compared to hypoxemia, which could indicate that hypercapnia had a greater effect on the heart. To investigate whether the changes were caused by the pressure or the HR, the Fridericia corrected R-Tend intervals were explored further.

2) *R-Tend corrected interval*: RTc shows no significant difference for the interventions compared to the baseline. Figure 10 shows how the RTc intervals decreases when the RV systolic pressure increases. Though, the intervals decreases for the specific subject it is not a consistent result throughout subjects. This is also reflected in the 95% confidence interval. Although, the RTc intervals does not significantly change during the interventions compared to the baselines, it requires further investigation in order to determine Tend to be a stationary reference point.

3) *S2onset-Tend interval*: The results from table VII presents a significant change in the occurrence of the S2onset for hypercapnia compared to the baseline relative to Tend. The table shows a negative 95% confidence interval during hypercapnia indicating that either the HR or the RV systolic pressure has an effect on a premature S2onset. As seen in table III the HR increases for both interventions, but only hypercapnia shows a significant difference in the S2onset-Tend intervals compared to the baseline. This suggest that the HR is not the primary effect on the changes in the intervals. The table shows a higher increase in RV systolic pressure during hypercapnia, which therefore can be the main cause of the change.

B. Study limitations

1) *Detection method*: There is no standardized method to detect the attributes needed for the comparison of S2onset and Tend, therefore the used methods to find Tend and S2onset were developed. To ensure the precision of the detected points, a comparison to a gold-standard would have been conducted, to verify the developed methods. Since there is no gold-standard to verify the detected points, it could not be tested. This implicates the precision of the detected points and thus the accuracy of the performed comparisons. To improve the accuracy the developed methods should be tested thoroughly to ensure precise detection of Tend and S2onset. Even though there is no quantitative method to test the precision of the detections, plots of the ECG and SCG signals with the detections

TABLE IV

R-TEND STATISTICS FOR DIFFERENCES BETWEEN BASELINE TO HYPOXEMIA AND HYPERCAPNIA. A NEGATIVE DIFFERENCE INDICATES EARLIER DETECTIONS DURING INTERVENTION, WHILE A POSITIVE DIFFERENCE INDICATES LATER DETECTIONS.

R-Tend	Mean (ms)	Std. dev (ms)	Diff. from mean (ms)	Diff. Percentage	P-value	95% Confidence interval (ms)
Baseline before hypercapnia	331.8	6.8				
Hypercapnia	297.4	44.9	-34.3	10.4%	0.019	-6.6 to -62
Baseline before hypoxemia	341.3	6.5				
Hypoxemia	332.2	21.9	-9.1	2.9%	0.143	3.4 to -21.7

TABLE V

R-S2ONSET STATISTICS FOR DIFFERENCES BETWEEN BASELINE TO HYPOXEMIA AND HYPERCAPNIA. A NEGATIVE DIFFERENCE INDICATES EARLIER DETECTIONS DURING INTERVENTION, WHILE A POSITIVE DIFFERENCE INDICATES LATER DETECTIONS.

R-S2onset	Mean (ms)	Std. dev (ms)	Diff. from mean (ms)	Diff. Percentage	P-value	95% Confidence interval (ms)
Baseline before hypercapnia	303.7	17.5				
Hypercapnia	245.3	47.6	-58.4	19.2%	<0.001	-28,9 to -88.0
Baseline before hypoxemia	314.3	19.0				
Hypoxemia	296.1	36.0	-18.2	5.8%	0.046	-0.4 to -36.0

TABLE VI

RTC STATISTICS FOR DIFFERENCES BETWEEN BASELINE TO HYPOXEMIA AND HYPERCAPNIA. A NEGATIVE DIFFERENCE INDICATES EARLIER DETECTIONS DURING INTERVENTION, WHILE A POSITIVE DIFFERENCE INDICATES LATER DETECTIONS.

RTc	Mean (ms)	Std. dev (ms)	Diff. from mean (ms)	Diff. Percentage	P-value	95% Confidence interval (ms)
Baseline before hypercapnia	346.3	6.9				
Hypercapnia	324.7	16.9	-21.5	6.2%	0.206	13.2 to -56.3
Baseline before hypoxemia	356.5	6.8				
Hypoxemia	352.7	16.0	-3.8	1.1%	0.636	13.0 to -20.7

TABLE VII

S2ONSET-TEND STATISTICS FOR DIFFERENCES BETWEEN BASELINE TO HYPOXEMIA AND HYPERCAPNIA. A NEGATIVE DIFFERENCE INDICATES EARLIER DETECTIONS DURING INTERVENTION, WHILE A POSITIVE DIFFERENCE INDICATES LATER DETECTIONS.

S2onset-Tend	Mean (ms)	Std. dev (ms)	Diff. from mean (ms)	Diff. Percentage	P-value	95% Confidence interval (ms)
Baseline before hypercapnia	-28.0	19.1				
Hypercapnia	-59.9	28.8	-32.0	114.4%	<0.001	-18.7 to -45.3
Baseline before hypoxemia	-27.0	20.4				
Hypoxemia	-43.1	28.4	-16.1	59.6%	0.089	2.8 to -35

were looked through manually. Examples of the detections can be seen in appendix II.

2) Exclusion criteria: In some cases a biphasic T-wave was observed. These were excluded since the developed model only looked at positive T-waves to find T-peak and Tend. To prevent these from being excluded, a function that accounts for biphasic T-waves could be developed.

The developed method used the P-wave to find Tend, though it was seen with a high HR that the P-wave was hidden in the T-wave. In those cases, Tend could not be found using the P-wave. Therefore another method, not using the P-wave, would be more sufficient.

In some datasets the combinations of high HRs and late T-waves were observed. The function of late T-wave made the search window for T-peak longer, but since the RR interval got shorter with a higher HR, the searching windows would overlap. Therefore, a method combining a short RR interval window and a late T-peak window would optimize the amount of detections and the precision of the results.

C. Data collection from pigs

The recordings from the subjects could last up to 3 hours, where they were induced with the interventions multiple times. This multiple burden may have caused some stress reactions on the subjects, that made the results from later recordings more unreliable. Some of the differences between the earlier and later recordings of the same subject, were changes in HR, differences in ECG, and some

subjects got ventricular fibrillation. These factors contributed to many excluded cases.

Multiple baselines were measured on each subject as a reference, which visualized the current condition of the subjects. One baseline recording had several intervention recordings associated. This resulted in a comparison of a baseline and intervention multiple hours apart which was not optimal. A baseline recording before each intervention recording would have optimized the precision of the changes from baseline to intervention.

To simulate PH the subjects were induced with hypoxemia and hypercapnia. As observed in table III, the systolic pressure in the RV increases to above 25 mmHg, thereby matching the pressure during PH, as described in section I. Therefore making hypoxemia and hypercapnia reliable methods to induce acute PH. The hypoxemia and hypercapnia recordings were in an acute state of PH, thus the results do not reflect a chronic state of PH. In table III the RV systolic pressure increases by 74.8% during hypercapnia and 31.8% during hypoxemia, which could be compared to a more chronic state of PH. However, the effect chronic PH has on S2onset can not be deducted by the results.

Because the experiments were executed on pigs it can not directly transfer to humans, but in general the anatomy of a pigs heart has many similarities to a human heart, such as size, valves and heart chambers. This suggests that their physiological responses can be a reliable model for how the human cardiovascular system might respond under similar conditions.

D. Future perspective

This study leads to further research of how S2 is affected by an increased pressure on the pulmonic valve. For this to be possible, a correction model for the timing of S2onset at different HRs should be made, to tell whether the difference was influenced by HR or the pressure. This could be investigated by conducting experiments on the pigs, by raising their HR and investigating how this affects the S2onset-Tend interval. A standard for detection of S2onset and Tend is needed, for this study to be comparable to similar studies. The amount of test subjects should be increased for a more generalizable result. To transfer the results of S2onset and Tend comparison onto humans, other tests must be carried out.

V. CONCLUSION

It can be concluded from this study that there is a significant deviation of S2onset-Tend interval when the pigs got PH induced by hypercapnia. The results from hypoxemia induced PH showed a tendency, but did not show a significant deviation of S2onset-Tend from baseline to intervention. Before SCG and ECG in combination as methods to diagnose PH, further research is required.

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APPENDIX I SIGNAL PROCESSING

To compare the T_{end} and the $S2_{onset}$, the ECG and SCG signals must be filtered so the P-, R- and T- wave in the ECG were possible to detect. Likewise, the SCG signal must be filtered to ensure that S2 was detectable.

1) *Downsampling*: ECG and SCG signals were downsampled from 5000 Hz to 500 Hz. This alteration was made to decrease the compiling time of the script. 500 Hz was a sampling frequency that satisfied the Nyquist theorem, since the highest relevant frequencies from the signals was 140 Hz, which resulted in a minimum sampling frequency at 280 Hz.

A. ECG filtering

The ECG filtering was designed to remove noise and unwanted frequency components, while preserving the attributes of a traditional ECG. The goal was to remove the following components from the ECG:

- 1) Powerline interference caused by electrical power systems.
- 2) High-frequency noise caused by muscle activity.
- 3) Low-frequency noise caused by respiration.

A frequency analysis in form of a CWT, seen in figure 12, showed a defined 50 Hz frequency caused by the powerline interference.

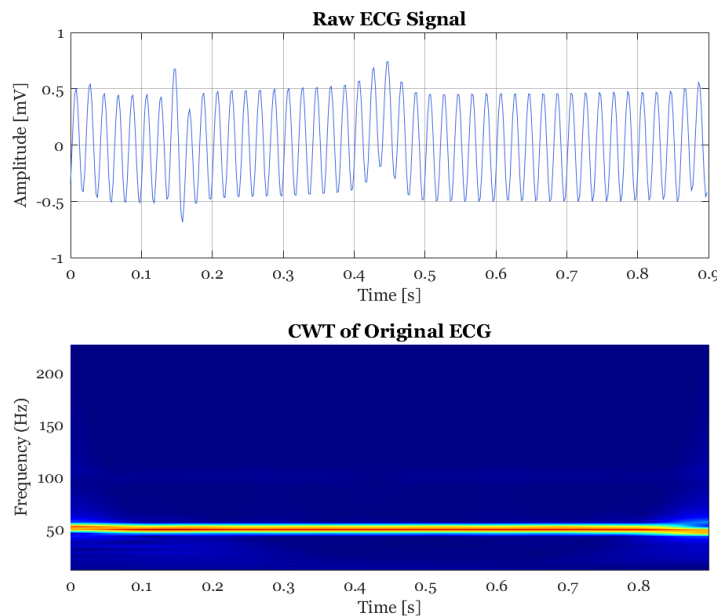


Fig. 12. The ECG signal in the top plot and the corresponding CWT in the bottom plot

In order to remove the powerline interference a Low pass FIR filter was implemented. The filter had a cutoff frequency of 40 Hz, to remove as much of the noise as possible without removing the frequencies describing the ECG. A FIR filter was chosen since it has a linear phase response, which was an important factor considering the study wanted to compare two timestamps. Similarly, the `'filtfilt'` function from MatLab was also used to ensure zero-phase filtering and therefore filtering the signal both forward and backwards. The effect of the filter can be seen in figure 13.

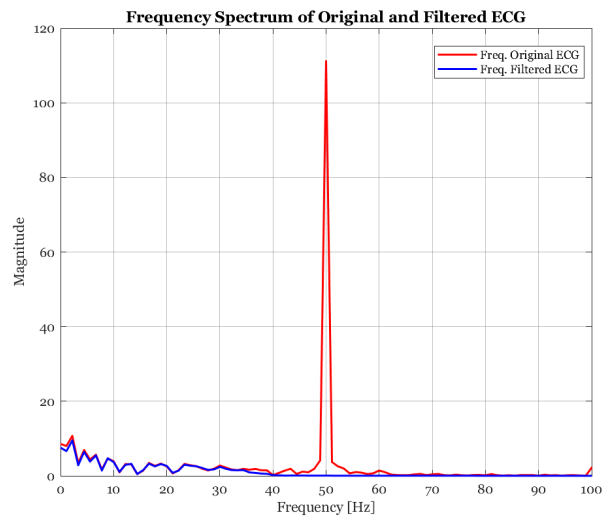


Fig. 13. The ECG signal in the frequency domain, is shown as a FFT plot, where the red signal is the unfiltered ECG signal and the green signal is the same ECG signal after applying the 40Hz lowpass filter.

To ensure that the low-frequency noise did not disturb the baseline of the signal, a High pass FIR filter with a cutoff frequency of 0.5 Hz and a filter order of 70 was applied. The result of the filtered ECG signal can be seen in figure 14.

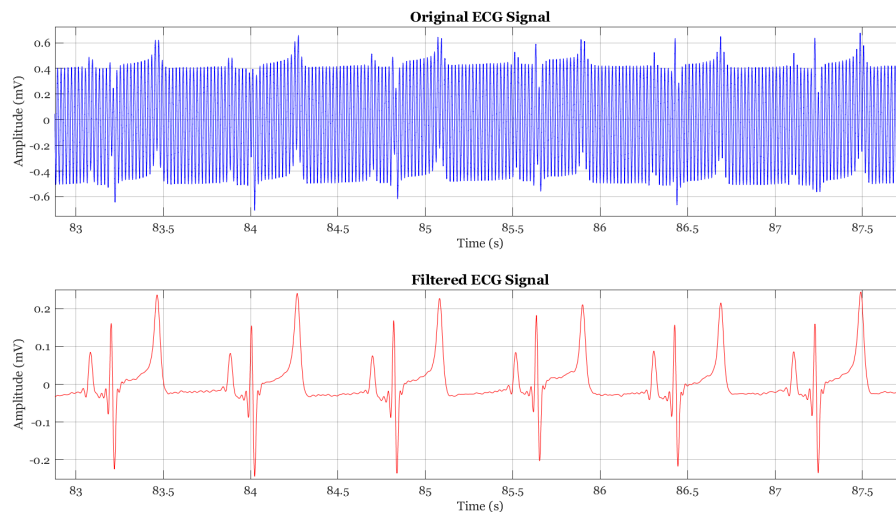


Fig. 14. In the first plot is the unfiltered ECG signal. In the lower plot is the filtered ECG signal

B. ECG event detection

In order to compare the timing of the S2onset and Tend, it was necessary to detect R-, T- and P-waves and from these detect Tend in the ECG signal.

1) *Detection of R-peak*: The R-peak was isolated by investigating the signal in a time frequency analysis using a CWT. The CWT can be seen in the second subplot in figure 15. Based on the wavelet analysis, a bandpass filter with cutoff frequencies at 14 Hz to 40 Hz were chosen to isolate the R-wave. To eliminate phase shifting in time a bandpass FIR filter was chosen, and applied with the '*filtfilt*' function on the filtered ECG signal.

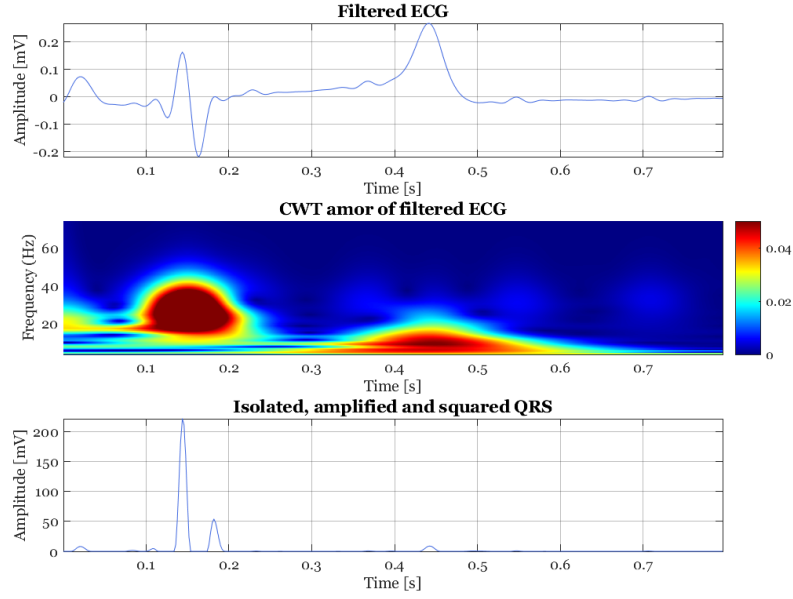


Fig. 15. The steps to isolate the R-peak is shown. The upper plot shows the filtered ECG signal which the isolation of the QRS-complex is based on. In the middle plot the CWT analysis is shown, which the QRS-complex isolation filter design is based on. In the lower plot the result from the isolation of the QRS-complex is presented.

The filtering process enhanced the visibility of the QRS-complex. To improve the clarity of the R-wave and ensure that the set to zero as shown in formula 4. To further highlight the R-wave, the signal was squared to emphasize the differences in peak magnitude as in formula 5. The processed signal can be seen in the third plot in figure 15. The MatLab function `'findpeaks'` was used, to find the R-peaks, which was defined as peaks with a peak height at minimum 0.01 mV and a window of at least 0.2 s.

$$QRS_{nonzero}(t) = \begin{cases} ECG(t), & \text{if } ECG(t) \geq 0 \\ 0, & \text{if } ECG(t) < 0 \end{cases} \quad (4)$$

$$QRS_{amplified}(t) = QRS_{nonzero}(t)^2 \quad (5)$$

2) Detection of T-peak and P-peak: To detect the T-peak, the R-peaks are used as reference points. For each RR-interval, a T-wave was expected to occur within a window of 0.15 to 0.35 s after the R-peak. Within this window, the maximum value was identified as the T-peak. The P-peak was detected with a similar method as the T-peak. For each RR-interval, a P-wave was expected to occur within a window of 0.7 to 0.17 s before the R-peak. In figure 16 the detected points are presented with the search points.

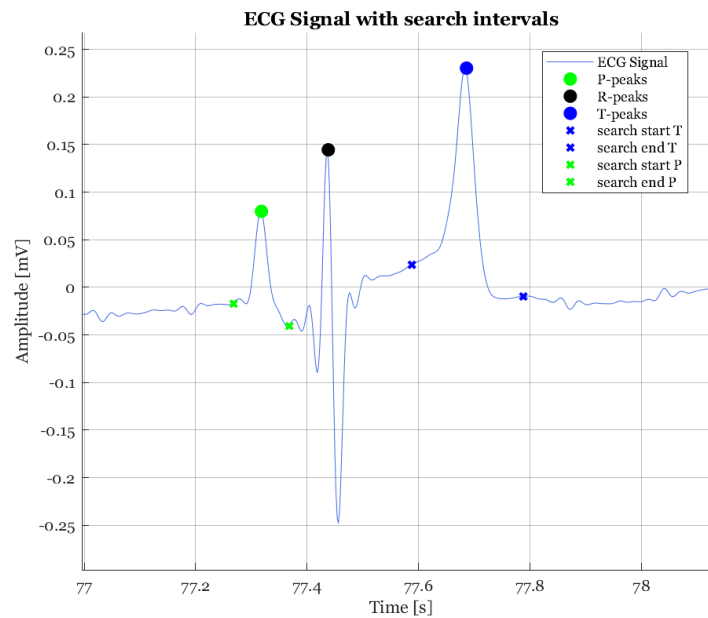


Fig. 16. The ECG signal with marked search window edges for detecting the peak of the P and T wave. The X's visualize where the search starts and stops to either locate the T- or P-peak.

The original function for detecting the ECG events was developed based on the baseline recordings. After applying the algorithm on the intervention recordings, some issues occurred. The most common was that the HR increased significantly, which caused problems with the R-peak detection. Another issue was that the T-wave occurred later than normal, which caused problems with the Tend detection. Additional functions were developed to handle these issues, the variations in the functions are seen in table VIII. The original function would skip R-peaks if the RR-interval was shorter than 0.52 s, equivalent to 115 bpm, to ensure double beats were ignored. This was handled by using a shorter RR-interval accepting a HR up to 200 bpm. In this function, the search intervals for P and T were also adjusted. The search intervals would overlap if the HR exceed 150 bpm, which could cause a wave peak to be detected as both P- and T- peaks. In these cases the peaks were excluded. In cases where the T-wave occurred later than normal, the search interval for the T-wave was expanded.

TABLE VIII

TABLE OF ALTERATIONS IN DIFFERENT FUNCTIONS TO DETECT ATTRIBUTES IN THE ECG

Function	R-peak Window	T-peak Window	P-peak Window
Original Function	RR > 0.52 s (115 bpm)	+0.15 to +0.35	-0.17 to -0.07
High Heart Rate	RR > 0.30 s (200 bpm)	+0.10 to +0.25	-0.15 to -0.05
Delayed T-wave	RR > 0.52 s (115 bpm)	+0.25 to +0.55	-0.17 to -0.07

3) *Detection of Tend*: To detect Tend the detected attributes in the ECG were utilized as visualized in figure 17.

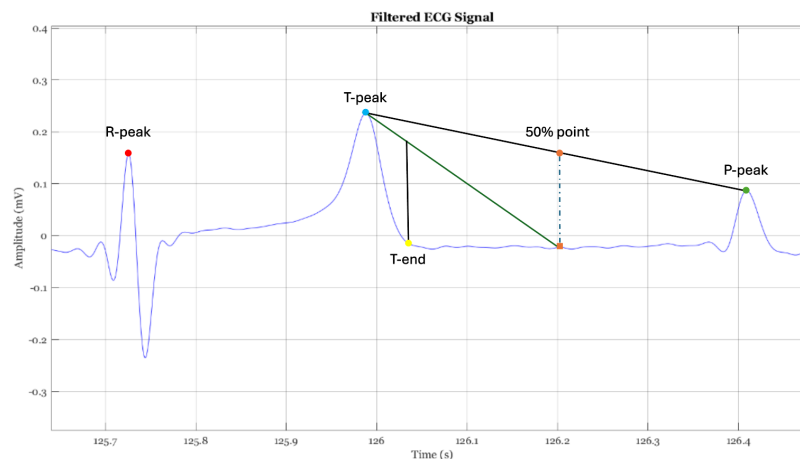


Fig. 17. Illustration of the method to find the T-end spot

From the T-wave to the P-wave in the same RR-interval, an isoelectrical line was expected. Therefore, the point in the middle of the

T-peak and the P-peak was found as the 50% point. From the 50% point on the isoelectrical line, a green line to the T-peak point was drawn. Tend was detected by finding the point on the ECG signal which had the largest vertical distance to the green line.

C. Detection of S2

In the SCG signal in figure 18 two oscillations were observed, one for S1 when the mitral and tricuspid valve closes during the R-wave on the ECG. The S2 sound was the sound which occurred near the T-wave and when the pulmonary and aortic valve closes.

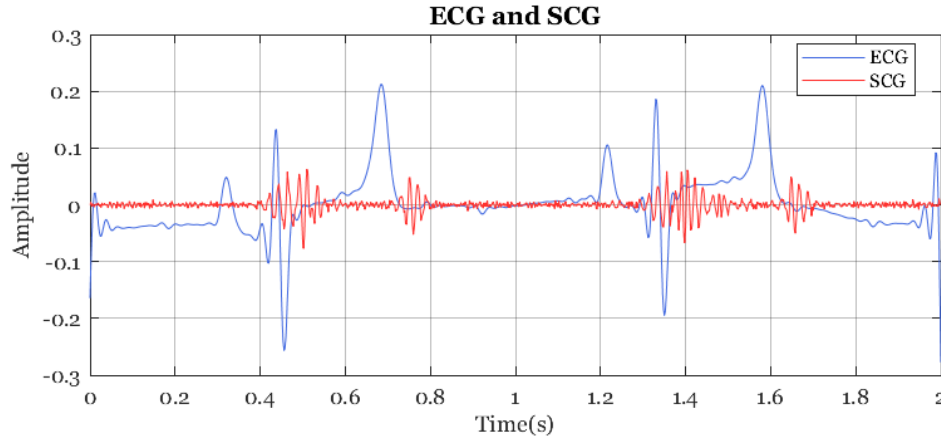


Fig. 18. ECG and SCG in the same figure

1) *Filtering of SCG signal*: The purpose of filtering SCG signals was to remove noise in between S1 and S2. In figure 19 the SCG-signal and a corresponding CWT shows, that the signal has energy for frequencies below 40 Hz continuously throughout the signal.

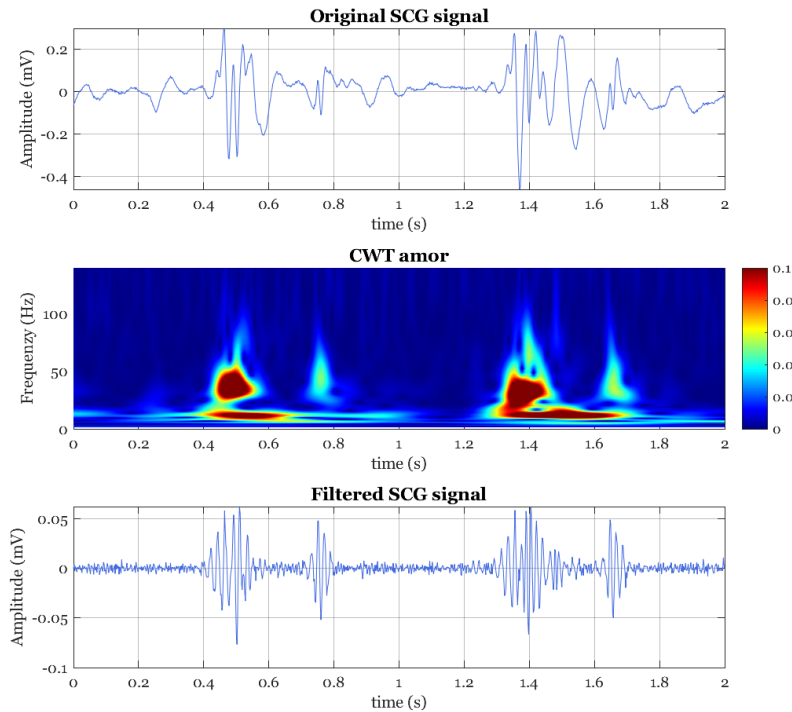


Fig. 19. The top figure shows the raw SCG signal and the bottom figure shows a CWT of it.

In figure 19 a highpass filter was applied which removed the frequencies below 40 Hz. This alteration of the signal showed S1 and S2 clearly from the rest of the signal. When doing this filtration, it was important to use a filter that did not alter the timing of S1 and S2 in the signal. Therefore, a FIR filter with a hamming window was used, since it reduced the leakage of the signal, and did not shift the timing of the signal. The cutoff frequency of the signal was 40 Hz, and the filter order was set to 70.

2) *Onset of S2 detection*: To detect the onset of S2 an envelope signal of the SCG's Shannon energy was made. A MatLab function "sigrangebinmask" was used on the envelope, this function detected the heart sounds. The function created an array of binary operators that marks 1 when the signal reached a height of 0.3% of the maximum value in the signal for 45 ms. To avoid any murmurs being detected as heart sounds, a for-loop was created to look through the maximum values in the intervals of the marks within the array. Additionally, the binary values were set to zero when it did not reach 1% of the maximum signal. To mark the start of the wave, the first point of each interval in the array was added as a marker on the signal. To distinguish between S1 and S2, the detected R-peak in the ECG was used, since it was observed that the S1 always occurs after the R-wave. The midpoint between R- and T-peak in the ECG was used as a searching point for S2 in the SCG. The first mark after the searching point in the envelope signal was found. The heart sound after this mark was S2. The onset of S2 is found with a 0.3 s window to look forward in, where S2onset was defined when the signal differs from the zero- line.

D. Compare S2 and T-end occouring time

The detected events, P-, R-, T-peak, and Tend in the ECG and the S2onset in the SCG was stored in an array. This was to ensure that the correct S2 occurred within the same RR-interval. If a value of one or more events in the ECG or SCG signal were not found, the entire row was removed. The calculation of the deviations between S2onset and Tend was made by:

$$Deviation = S2_{onset} - T_{end}$$

If the result was negative, it indicated that S2onset occurred before Tend. The comparison was made on both a baseline and an intervention recording, and the mean and standard deviation for both was found. The difference in the mean was found by:

$$Difference = \mu_{Baseline} - \mu_{Intervention}$$

If the results were negative, it indicated that $\mu_{Intervention}$ occurred earlier than $\mu_{Baseline}$.

E. Deviations in QT-interval

Since the Tend point was used as a reference, it was important to investigate if the points stayed constant in both the baseline and the intervention recordings. If the Tend reference point deviated it could result in undependable results of the S2onset and Tend deviations. Therefore, the QT-intervals were investigated in both the baseline and the intervention signals. In Figure 20 the mean of the baseline and intervention beats were aligned around the R-peak. The figure showed a visual difference in the QT-interval when the pulse differentiated.

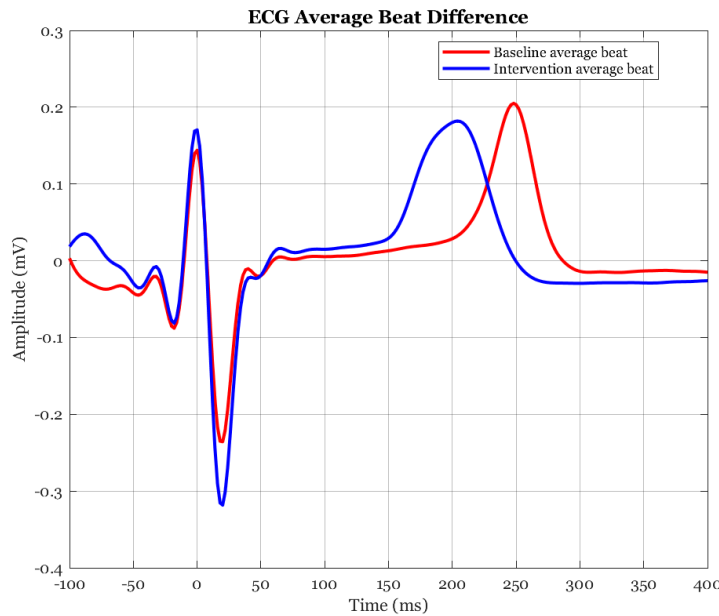


Fig. 20. Average cardiac cycle during baseline and intervention centered around the R-peak.

To compare signals with two different heart rates the Fridericia correction formula was used:

$$QT_c = \frac{QT}{\sqrt[3]{RR}}$$

Where QT is the interval from Q-wave to T-wave. In this study the interval was calculated from the R-peak to T-end as well as the RR -interval. The mean and standard deviation for both the baseline and the intervention signal was found, and the difference in the means was calculated.

F. Exclusion criteria

To ensure the results gained from the comparisons are accurate the following exclusion criteria were made:

- | | |
|-------------------------------------|----------------------------|
| 1) Baseline detected beats < 50 | 4) Late T-wave and high HR |
| 2) Intervention detected beats < 50 | 5) Missing signal |
| 3) Biphasic T-wave | 6) P-wave hidden in T-wave |

The baseline and intervention detected beats were arrays contained datapoints for both the ECG and SCG. Thus, exclusion criteria 1) and 2) were to ensure enough data points to calculate a valid mean during the baseline and the intervention. Subacute ischemia can in the early stages create a biphasic T-wave. [25]. This can effect the detection of T-end, which is the reason for exclusion criteria 3). Three different functions to find T-peak and T-end were made, based on a normal heart rate, a HR exceeding 115 bpm and a late T. The development of a function combining both the HR exceeding 115 bpm and the late T-wave, was de-prioritized, since the RR interval gets shorter with a higher HR making the search windows for the T- and P-wave overlap. This resulted in exclusion criteria 4). To be able to make the comparisons with the ECG and SCG, readable signals were required. To ensure this, the datasets which had missing signals in either ECG or SCG were excluded resulting in exclusion criteria 5). When the HR exceeds 100 bpm, tachycardia is reached. A higher HR could result in a hidden P-wave in the T-wave [26]. This affects the detection of the P-wave, and with that the T-wave, which resulted in the exclusion criteria 6).

APPENDIX II DATA PROTOCOL

To ensure conformity in how the data was processed, several plots were checked for valid detections. To check the HR of the subjects, the monitor data was viewed, to check if the function needed to be changed to high HR or late T-wave. Beside this, the systolic pressure and oxygen saturation were looked through, to check, that the highest systolic pressure was where the oxygen saturation fell.

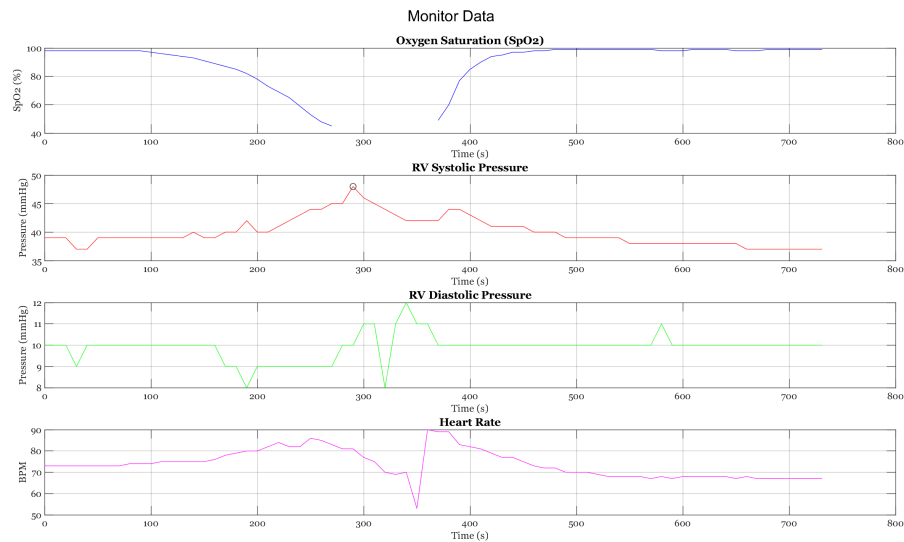


Fig. 21. Monitor data for subject G10

Figure 21 presents the monitor data for subject G10. The HR, systolic and diastolic pressure, and the oxygen saturation of the subject can be viewed. To decide whether the datasets were included or excluded, the baseline and intervention detections were checked for correct detections. Examples of these are seen in figure 23 for baseline and figure 22 for the intervention.

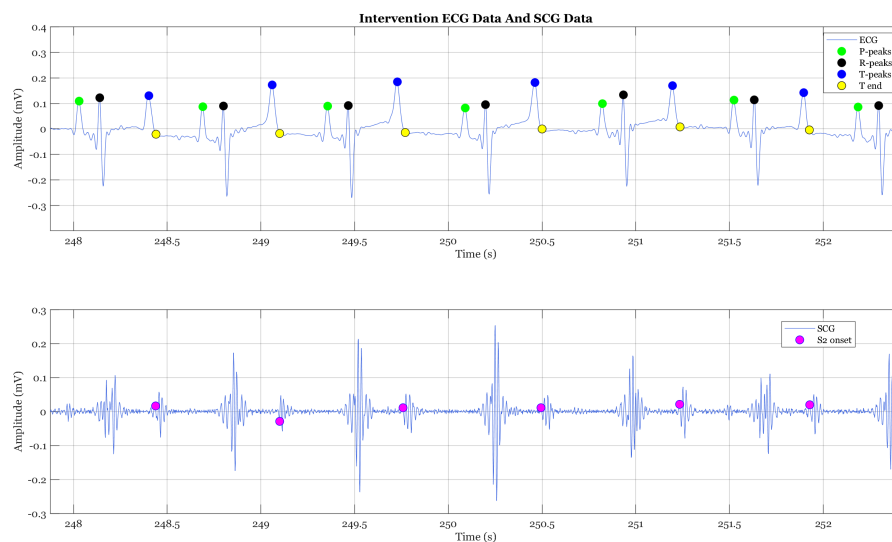


Fig. 22. Detections of Tend and S2onset during Intervention recording.

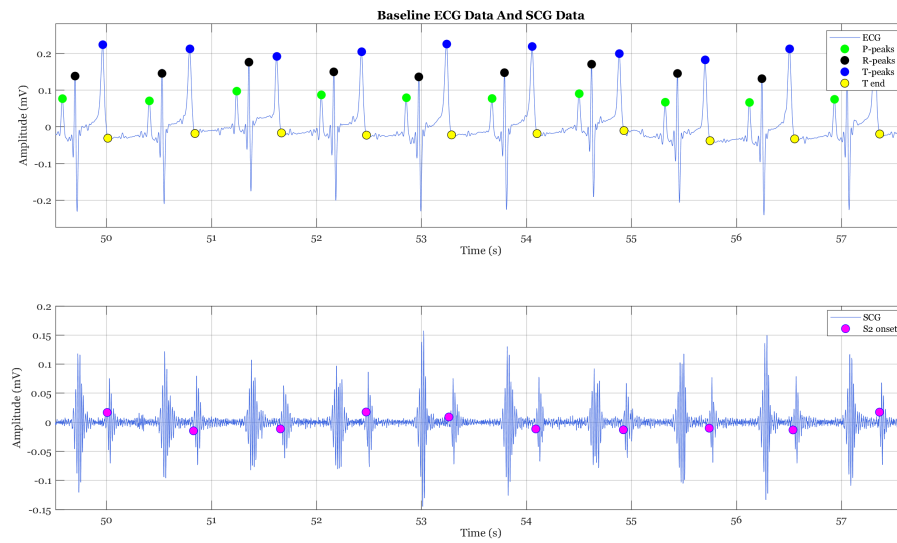


Fig. 23. Detections of Tend and S2onset during Baseline recording.

The baseline and intervention arrays were checked for correct detections, if there were less than 50 correct detections, the dataset was excluded.

After having included a dataset, the mean values for baseline and intervention for the following intervals, were noted in an excel ark:

- S2onset-Tend
- RTc
- R-Tend
- R-S2onset

APPENDIX III RESULTS

Pig	S2onset-Tend		R-Tend		RTc		R-S2onset	
	Baseline	Hypercapnia	Baseline	Hypercapnia	Baseline	Hypercapnia	Baseline	Hypercapnia
G2_0857_hypercapnia_onlySCG	-0,045540541	-0,043987097	0,270493151	0,278023358	0,323446614	0,278804885	0,224849315	0,227159124
G2_1139_hypercapnia_onlySCG	-0,002891892	-0,029253165	0,294712329	0,298011416	0,281434081	0,338693969	0,292054795	0,294075342
G2_1208_hypercapnia_onlySCG	0,011027027	-0,028819876	0,286888889	0,286405028	0,274845068	0,338663899	0,298166667	0,275701117
G2_1245_hypercapnia_onlySCG	0,01112	-0,009015075	0,290181818	0,282704545	0,285388624	0,284321896	0,301252525	0,262106643
G4_1003_hypercapnia_onlySCG	-0,04384	-0,03924	0,342909091	0,345414918	0,367346433	0,348002061	0,299272727	0,298941725
G5_0902_hypercapnia_onlySCG	-0,053153846	-0,116402299	0,348629834	0,352302302	0,400562179	0,479216287	0,295414365	0,268520521
G5_0928_hypercapnia_onlySCG	-0,053153846	-0,1216125	0,348629834	0,371004348	0,400562179	0,474769007	0,295414365	0,265
G7_1042_hypercapnia_onlySCG	-0,049344	-0,051992701	0,361790323	0,267270962	0,389681205	0,294685328	0,312387097	0,206559576
G8_1413_hypercapnia_onlySCG	-0,031652968	-0,042515942	0,366550459	0,238030534	0,364676017	0,257257018	0,334862385	0,188145732
G9_0923_hypercapnia_onlySCG	-0,036835443	-0,040958904	0,359765079	0,229080689	0,386047183	0,260390946	0,322984127	0,182866727
G9_1000_hypercapnia_onlySCG	-0,036835443	-0,072378855	0,362216867	0,33618022	0,386047183	0,343650688	0,322984127	0,264137363
G10_0909_hypercapnia_onlySCG	-0,009714286	-0,062614679	0,296014388	0,290504578	0,307184623	0,286609803	0,286258993	0,246923703
G10_0948_hypercapnia_onlySCG	-0,009714286	-0,055264574	0,296014388	0,28466343	0,307184623	0,301232422	0,286258993	0,257775081
G10_1045_hypercapnia_onlySCG	-0,009724138	-0,065144144	0,310417391	0,288166365	0,331758773	0,293012274	0,300747826	0,266495479
G11_1256_hypercapnia_onlySCG	-0,043687075	-0,071928571	0,386534247	0,275474442	0,367292715	0,272100944	0,343082192	0,176362851
G11_1345_hypercapnia_onlySCG	-0,043687075	-0,108044944	0,386534247	0,33562963	0,367292715	0,34454697	0,343082192	0,243597884
Mean	-0,027976738	-0,059948333	0,331767646	0,297429173	0,346296888	0,3247474	0,303692043	0,245273054
Difference		0,031971595		0,034338473		0,021549489		0,058418989

Fig. 24. The table shows the results for the calculated means for each pig during baseline and hypercapnia. The mean of the means was calculated as well as the difference from the baseline to the intervention.

Pig	S2onset-Tend		R-Tend		RTc		R-S2onset	
	Baseline	Hypoxemia	Baseline	Hypoxemia	Baseline	Hypoxemia	Baseline	Hypoxemia
G2_1125_hypoxemia_onlySCG	-0,014732984	-0,027303371	0,297473684	0,292703325	0,304019715	0,319079512	0,28268421	0,27704092
G2_1153_hypoxemia_onlySCG	-0,002891892	-0,004836364	0,294712329	0,274940276	0,281434081	0,291234841	0,2920548	0,27522205
G2_1229_hypoxemia_onlySCG	0,028568182	-0,017737705	0,261816092	0,277861035	0,265816627	0,302304432	0,29048276	0,2826158
G4_0925_hypoxemia_onlySCG	-0,04384	-0,014137931	0,342909091	0,322150144	0,367346433	0,326735679	0,29927273	0,29763811
G5_0915_hypoxemia_onlySCG	-0,053153846	-0,054539216	0,348629834	0,38928172	0,400562179	0,418682994	0,29541437	0,33095914
G5_0951_hypoxemia_SCG+PCG	-0,049145455	-0,038313725	0,407504587	0,388330097	0,416421981	0,424809089	0,35818349	0,35161812
G6_0939_hypoxemia_onlySCG	-0,041565891	-0,026357895	0,354295337	0,347777497	0,373875126	0,367109183	0,31270985	0,31751454
G7_1025_hypoxemia_onlySCG	-0,049344	-0,020629213	0,361790323	0,363115005	0,389681205	0,380884474	0,3123871	0,32398905
G8_1346_hypoxemia_onlySCG	-0,031652968	-0,024168675	0,366550459	0,349193948	0,364676017	0,370901436	0,33486239	0,31472902
G8_1534_hypoxemia_SCG+PCG	-0,02405042	-0,103866667	0,365508475	0,329810462	0,415319455	0,358391665	0,34140678	0,26050644
G9_0946_hypoxemia_onlySCG	-0,036835443	-0,020033613	0,359765079	0,291418719	0,386047183	0,315679306	0,32298413	0,23211823
G9_1104_hypoxemia_onlySCG	-0,02405042	-0,103866667	0,365508475	0,329810462	0,415319455	0,358391665	0,34140678	0,26050644
G10_0855_hypoxemia_onlySCG	-0,009714286	-0,027213333	0,296014388	0,292543066	0,307184623	0,313632741	0,28625899	0,28575183
G10_0936_hypoxemia_onlySCG	-0,009714286	-0,038744186	0,296014388	0,294694545	0,307184623	0,322785055	0,28625899	0,27902909
G10_1033_hypoxemia_onlySCG	-0,009724138	-0,019422819	0,310417391	0,310522427	0,331758773	0,334313866	0,30074783	0,30230079
G11_1241_hypoxemia_onlySCG	-0,043687075	-0,065763889	0,386534247	0,390233766	0,367292715	0,385493178	0,34308219	0,33174315
G11_1329_hypoxemia_onlySCG	-0,043687075	-0,125964602	0,386534247	0,402487805	0,367292715	0,405629809	0,34308219	0,31058537
Mean	-0,027013059	-0,043111757	0,341292849	0,332169076	0,356543112	0,352709349	0,31431056	0,29610989
Difference		0,016098698		0,009123772		0,003833764		0,01820067

Fig. 25. The table shows the results for the calculated means for each pig during baseline and hypoxemia. The mean of the means was calculated as well as the difference from the baseline to the intervention.

Baseline			Hypercapnia			Comparison	
Pig	Mean	Std. dev	Pig	Mean	Std. dev	Difference	p-value
G2_0852_baseline_onlySCG	0.270493151	0.001519533	G2_0857_hypercapnia_onlySCG	0.278023358	0.112730489	-0.007530207	<0.005
G2_1137_baseline_onlySCG	0.294712329	0.006858651	G2_1139_hypercapnia_onlySCG	0.298011416	0.025165325	-0.003299087	<0.005
G2_1207_baseline_onlySCG	0.286888889	0.005543479	G2_1208_hypercapnia_onlySCG	0.286405028	0.034937998	0.000483861	<0.005
G2_1243_baseline_onlySCG	0.290181818	0.005406654	G2_1245_hypercapnia_onlySCG	0.282704545	0.026688773	0.007477273	<0.005
G4_0923_baseline_onlySCG	0.342909091	0.008321463	G4_1003_hypercapnia_onlySCG	0.345414918	0.043018954	-0.002505828	<0.005
G5_0845_baseline_onlySCG	0.348629834	0.00211371	G5_0902_hypercapnia_onlySCG	0.352302302	0.048424649	-0.003672468	<0.005
G5_0845_baseline_onlySCG	0.348629834	0.00211371	G5_0928_hypercapnia_onlySCG	0.371004348	0.05280366	-0.022374514	<0.005
G7_0855_baseline_onlySCG	0.361790323	0.016363391	G7_1042_hypercapnia_onlySCG	0.267270962	0.034608791	0.094519361	<0.005
G8_1341_baseline_onlySCG	0.366550459	0.030216267	G8_1413_hypercapnia_onlySCG	0.238030534	0.047487883	0.128519924	<0.005
G9_0900_baseline_onlySCG	0.359765079	0.005229332	G9_0923_hypercapnia_onlySCG	0.229080689	0.029007037	0.13068439	<0.005
G9_0900_baseline_onlySCG	0.362216867	0.006453268	G9_1000_hypercapnia_onlySCG	0.33618022	0.02931804	0.026036648	<0.005
G10_0849_baseline_onlySCG	0.296014388	0.003331852	G10_0909_hypercapnia_onlySCG	0.290504578	0.044489933	0.005509811	<0.005
G10_0849_baseline_onlySCG	0.296014388	0.003331852	G10_0948_hypercapnia_onlySCG	0.28466343	0.031049132	0.011350958	<0.005
G10_1031_baseline_onlySCG	0.310417391	0.003598203	G10_1045_hypercapnia_onlySCG	0.288166365	0.0302804	0.022251026	<0.005
G11_1236_baseline_onlySCG	0.386534247	0.004190571	G11_1256_hypercapnia_onlySCG	0.275474442	0.061689823	0.11105982	<0.005
G11_1236_baseline_onlySCG	0.386534247	0.004190571	G11_1345_hypercapnia_onlySCG	0.33562963	0.065961959	0.050904617	<0.005
Total mean	0.331767646	0.006798907		0.297429173	0.044853928	0.034338473	

Fig. 26. R-Tend results from baseline and hypercapnia including mean, standard deviation, difference and p-value.

Baseline			Hypoxemia			Comparison	
Pig	Mean	Std. dev	Pig	Mean	Std. dev	Change	p-value
G2_1121_baseline_onlySCG	0.297473684	0.007886553	G2_1125_hypoxemia_onlySCG	0.292703325	0.010287616	0.00477036	<0.005
G2_1137_baseline_onlySCG	0.294712329	0.006858651	G2_1153_hypoxemia_onlySCG	0.274940276	0.007735928	0.01977205	<0.005
G2_1225_baseline_onlySCG	0.261816092	0.006837871	G2_1229_hypoxemia_onlySCG	0.277861035	0.009522704	-0.0160449	<0.005
G4_0923_baseline_onlySCG	0.342909091	0.008321463	G4_0925_hypoxemia_onlySCG	0.322150144	0.026999903	0.02075895	<0.005
G5_0845_baseline_onlySCG	0.348629834	0.00211371	G5_0915_hypoxemia_onlySCG	0.38928172	0.014426391	-0.0406519	<0.005
G5_0949_baseline_SCG+PCG	0.407504587	0.001494582	G5_0951_hypoxemia_SCG+PCG	0.388330097	0.015940059	0.01917449	<0.005
G6_0932_baseline_onlySCG	0.354295337	0.001767184	G6_0939_hypoxemia_onlySCG	0.347777497	0.014996406	0.00651784	<0.005
G7_0855_baseline_onlySCG	0.361790323	0.016363391	G7_1025_hypoxemia_onlySCG	0.363115005	0.026604744	-0.0013247	<0.005
G8_1341_baseline_onlySCG	0.366550459	0.030216267	G8_1346_hypoxemia_onlySCG	0.349193948	0.070655564	0.01735651	<0.005
G8_1518_baseline_onlySCG	0.365508475	0.002007447	G8_1534_hypoxemia_SCG+PCG	0.329810462	0.026553531	0.03569801	<0.005
G9_0900_baseline_onlySCG	0.359765079	0.005229332	G9_0946_hypoxemia_onlySCG	0.291418719	0.035432785	0.06834636	<0.005
G9_1102_baseline_onlySCG	0.365508475	0.002007447	G9_1104_hypoxemia_onlySCG	0.329810462	0.026553531	0.03569801	<0.005
G10_0849_baseline_onlySCG	0.296014388	0.003331852	G10_0855_hypoxemia_onlySCG	0.292543066	0.009592728	0.00347132	<0.005
G10_0849_baseline_onlySCG	0.296014388	0.003331852	G10_0936_hypoxemia_onlySCG	0.294694545	0.014218352	0.00131984	<0.005
G10_1031_baseline_onlySCG	0.310417391	0.003598203	G10_1033_hypoxemia_onlySCG	0.310522427	0.00588228	-0.000105	<0.005
G11_1236_baseline_onlySCG	0.386534247	0.004190571	G11_1241_hypoxemia_onlySCG	0.390233766	0.027652419	-0.0036995	<0.005
G11_1236_baseline_onlySCG	0.386534247	0.004190571	G11_1329_hypoxemia_onlySCG	0.402487805	0.0286628	-0.0159536	<0.005
Total mean	0.341292849	0.006455703		0.332169077	0.02186575	0.00912377	

Fig. 27. R-Tend results from baseline and hypoxemia including mean, standard deviation, difference an p-value

Baseline			Hypercapnia			Comparison	
Pig	Mean	Std. dev	Pig	Mean	Std. dev	Difference	p-value
G2_0852_baseline_onlySCG	0.224849315	0.036549689	G2_0857_hypercapnia_onlySCG	0.227159124	0.038940269	-0.002309809	<0.005
G2_1137_baseline_onlySCG	0.292054795	0.013465894	G2_1139_hypercapnia_onlySCG	0.294075342	0.009714497	-0.002020548	0.405792009
G2_1207_baseline_onlySCG	0.298166667	0.00609215	G2_1208_hypercapnia_onlySCG	0.275701117	0.035553959	0.022465549	<0.005
G2_1243_baseline_onlySCG	0.301252525	0.015514538	G2_1245_hypercapnia_onlySCG	0.262106643	0.049990646	0.039145882	<0.005
G4_0923_baseline_onlySCG	0.299272727	0.027626451	G4_1003_hypercapnia_onlySCG	0.298941725	0.038070362	0.000331002	<0.005
G5_0845_baseline_onlySCG	0.295414365	0.017661937	G5_0902_hypercapnia_onlySCG	0.268520521	0.052791052	0.026893844	<0.005
G5_0845_baseline_onlySCG	0.295414365	0.017661937	G5_0928_hypercapnia_onlySCG	0.265	0.044079251	0.030414365	<0.005
G7_0855_baseline_onlySCG	0.312387097	0.035183771	G7_1042_hypercapnia_onlySCG	0.206559576	0.047546678	0.10582752	<0.005
G8_1341_baseline_onlySCG	0.334862385	0.01195496	G8_1413_hypercapnia_onlySCG	0.188145732	0.065296532	0.146716653	<0.005
G9_0900_baseline_onlySCG	0.322984127	0.006827007	G9_0923_hypercapnia_onlySCG	0.182866727	0.04157694	0.1401174	<0.005
G9_0900_baseline_onlySCG	0.322984127	0.006827007	G9_1000_hypercapnia_onlySCG	0.264137363	0.045660612	0.058846764	<0.005
G10_0849_baseline_onlySCG	0.286258993	0.008213931	G10_0909_hypercapnia_onlySCG	0.246923703	0.060065623	0.03933529	<0.005
G10_0849_baseline_onlySCG	0.286258993	0.008213931	G10_0948_hypercapnia_onlySCG	0.257775081	0.049362228	0.028483912	<0.005
G10_1031_baseline_onlySCG	0.300747826	0.006078209	G10_1045_hypercapnia_onlySCG	0.266495479	0.055589579	0.034252347	<0.005
G11_1236_baseline_onlySCG	0.343082192	0.031226979	G11_1256_hypercapnia_onlySCG	0.176362851	0.064105512	0.166719341	<0.005
G11_1236_baseline_onlySCG	0.343082192	0.031226979	G11_1345_hypercapnia_onlySCG	0.243597884	0.063384066	0.099484308	<0.005
Total mean	0.303692043	0.017520336		0.245273054	0.047607988	0.058418989	

Fig. 28. R-S2onset results from baseline and hypercapnia including mean, standard deviation, difference and p-value.

Baseline			Hypoxemia			Comparison	
Pig	Mean	Std. dev	Pig	Mean	Std. dev	Change	p-value
G2_1121_baseline_onlySCG	0.282684211	0.028044848	G2_1125_hypoxemia_onlySCG	0.277040921	0.022895764	0.00564329	<0.005
G2_1137_baseline_onlySCG	0.292054795	0.013465894	G2_1153_hypoxemia_onlySCG	0.275222052	0.035243343	0.016832742	<0.005
G2_1225_baseline_onlySCG	0.290482759	0.044651587	G2_1229_hypoxemia_onlySCG	0.282615804	0.030432123	0.007866955	<0.005
G4_0923_baseline_onlySCG	0.299272727	0.027626451	G4_0925_hypoxemia_onlySCG	0.297638114	0.036820974	0.001634614	0.055945571
G5_0845_baseline_onlySCG	0.295414365	0.017661937	G5_0915_hypoxemia_onlySCG	0.33095914	0.029494639	-0.035544775	<0.005
G5_0949_baseline_SCG+PCG	0.358183486	0.025026714	G5_0951_hypoxemia_SCG+PCG	0.351618123	0.020955653	0.006565363	<0.005
G6_0932_baseline_onlySCG	0.312709845	0.013434739	G6_0939_hypoxemia_onlySCG	0.317514539	0.021974599	-0.004804694	<0.005
G7_0855_baseline_onlySCG	0.312387097	0.035183771	G7_1025_hypoxemia_onlySCG	0.323989047	0.02845578	-0.01160195	<0.005
G8_1341_baseline_onlySCG	0.334862385	0.01195496	G8_1346_hypoxemia_onlySCG	0.314729023	0.034595721	0.020133362	<0.005
G8_1518_baseline_onlySCG	0.34140678	0.006870272	G8_1534_hypoxemia_SCG+PCG	0.260506444	0.066316284	0.080900335	<0.005
G9_0900_baseline_onlySCG	0.322984127	0.006827007	G9_0946_hypoxemia_onlySCG	0.232118227	0.074764222	0.0908659	<0.005
G9_1102_baseline_onlySCG	0.34140678	0.006870272	G9_1104_hypoxemia_onlySCG	0.260506444	0.066316284	0.080900335	<0.005
G10_0849_baseline_onlySCG	0.286258993	0.008213931	G10_0855_hypoxemia_onlySCG	0.285751825	0.029424743	0.000507168	<0.005
G10_0849_baseline_onlySCG	0.286258993	0.008213931	G10_0936_hypoxemia_onlySCG	0.279029091	0.040398686	0.007229902	<0.005
G10_1031_baseline_onlySCG	0.300747826	0.006078209	G10_1033_hypoxemia_onlySCG	0.302300792	0.01604071	-0.001552965	<0.005
G11_1236_baseline_onlySCG	0.343082192	0.031226979	G11_1241_hypoxemia_onlySCG	0.331743146	0.032499851	0.011339046	<0.005
G11_1236_baseline_onlySCG	0.343082192	0.031226979	G11_1329_hypoxemia_onlySCG	0.310585366	0.024900779	0.032496826	<0.005
Total mean	0.314310562	0.018975205		0.296109888	0.035972481	0.018200674	

Fig. 29. R-S2onset results from baseline and hypoxemia including mean, standard deviation, difference and p-value.

Baseline			Hypercapnia			Comparison	
Pig	Mean	Std. dev	Pig	Mean	Std. dev	Difference	p-value
G2_0852_baseline_onlySCG	0.323446614	0.001255622	G2_0857_hypercapnia_onlySCG	0.278804885	0.012994439	0.044641729	<0.005
G2_1137_baseline_onlySCG	0.281434081	0.00652433	G2_1139_hypercapnia_onlySCG	0.338693969	0.016268081	-0.057259888	<0.005
G2_1207_baseline_onlySCG	0.274845068	0.005800107	G2_1208_hypercapnia_onlySCG	0.338663899	0.017380975	-0.063818831	<0.005
G2_1243_baseline_onlySCG	0.285388624	0.005902338	G2_1245_hypercapnia_onlySCG	0.284321896	0.006696867	0.001066728	0.179388444
G4_0923_baseline_onlySCG	0.367346433	0.009373751	G4_1003_hypercapnia_onlySCG	0.348002061	0.037845579	0.019344371	<0.005
G5_0845_baseline_onlySCG	0.400562179	0.00261833	G5_0902_hypercapnia_onlySCG	0.479216287	0.016965204	-0.078654107	<0.005
G5_0845_baseline_onlySCG	0.400562179	0.00261833	G5_0928_hypercapnia_onlySCG	0.474769007	0.037252234	-0.074206828	<0.005
G7_0855_baseline_onlySCG	0.389681205	0.016977742	G7_1042_hypercapnia_onlySCG	0.294685328	0.00963731	0.094995877	<0.005
G8_1341_baseline_onlySCG	0.364676017	0.02080552	G8_1413_hypercapnia_onlySCG	0.257257018	0.011267914	0.107418999	<0.005
G9_0900_baseline_onlySCG	0.386047183	0.005618459	G9_0923_hypercapnia_onlySCG	0.260390946	0.011640265	0.125656237	<0.005
G9_0900_baseline_onlySCG	0.386047183	0.005618459	G9_1000_hypercapnia_onlySCG	0.343650688	0.009133875	0.042396495	<0.005
G10_0849_baseline_onlySCG	0.307184623	0.003701396	G10_0909_hypercapnia_onlySCG	0.286609803	0.014947105	0.02057482	<0.005
G10_0849_baseline_onlySCG	0.307184623	0.003701396	G10_0948_hypercapnia_onlySCG	0.301232422	0.002185666	0.005952201	<0.005
G10_1031_baseline_onlySCG	0.331758773	0.004186584	G10_1045_hypercapnia_onlySCG	0.293012274	0.006972084	0.038746499	<0.005
G11_1236_baseline_onlySCG	0.367292715	0.007727409	G11_1256_hypercapnia_onlySCG	0.272100944	0.017892117	0.095191771	<0.005
G11_1236_baseline_onlySCG	0.367292715	0.007727409	G11_1345_hypercapnia_onlySCG	0.34454697	0.04179192	0.022745745	<0.005
Total mean	0.346296888	0.006884824		0.3247474	0.016929477	0.021549489	

Fig. 30. RTc results from baseline and hypercapnia including mean, standard deviation, difference and p-value.

Baseline			Hypoxemia			Comparison	
Pig	Mean	Std. dev	Pig	Mean	Std. dev	Change	p-value
G2_1121_baseline_onlySCG	0.304019715	0.008564889	G2_1125_hypoxemia_onlySCG	0.319079512	0.012758717	-0.015059797	<0.005
G2_1137_baseline_onlySCG	0.281434081	0.00652433	G2_1153_hypoxemia_onlySCG	0.291234841	0.00925475	-0.009800761	<0.005
G2_1225_baseline_onlySCG	0.265816627	0.007518479	G2_1229_hypoxemia_onlySCG	0.302304432	0.012107622	-0.036487805	<0.005
G4_0923_baseline_onlySCG	0.367346433	0.009373751	G4_0925_hypoxemia_onlySCG	0.326735679	0.018749375	0.040610754	<0.005
G5_0845_baseline_onlySCG	0.400562179	0.00261833	G5_0915_hypoxemia_onlySCG	0.418682994	0.005305234	-0.018120815	<0.005
G5_0949_baseline_SCG+PCG	0.416421981	0.00280291	G5_0951_hypoxemia_SCG+PCG	0.424809089	0.009423605	-0.008387108	<0.005
G6_0932_baseline_onlySCG	0.373875126	0.002591685	G6_0939_hypoxemia_onlySCG	0.367109183	0.016455549	0.006765943	<0.005
G7_0855_baseline_onlySCG	0.389681205	0.016977742	G7_1025_hypoxemia_onlySCG	0.380884474	0.012668632	0.008796731	<0.005
G8_1341_baseline_onlySCG	0.364676017	0.02080552	G8_1346_hypoxemia_onlySCG	0.370901436	0.009323288	-0.006225418	<0.005
G8_1518_baseline_onlySCG	0.415319455	0.002409795	G8_1534_hypoxemia_SCG+PCG	0.358391665	0.016437457	0.05692777	<0.005
G9_0900_baseline_onlySCG	0.386047183	0.005618459	G9_0946_hypoxemia_onlySCG	0.315679306	0.036328466	0.070367899	<0.005
G9_1102_baseline_onlySCG	0.415319455	0.002409795	G9_1104_hypoxemia_onlySCG	0.358391665	0.016437457	0.05692777	<0.005
G10_0849_baseline_onlySCG	0.307184623	0.003701396	G10_0855_hypoxemia_onlySCG	0.313632741	0.023571587	-0.006448119	<0.005
G10_0849_baseline_onlySCG	0.307184623	0.003701396	G10_0936_hypoxemia_onlySCG	0.322785055	0.023568468	-0.015600433	<0.005
G10_1031_baseline_onlySCG	0.331758773	0.004186584	G10_1033_hypoxemia_onlySCG	0.334313866	0.009869058	-0.002555093	0.008996104
G11_1236_baseline_onlySCG	0.367292715	0.007727409	G11_1241_hypoxemia_onlySCG	0.385493178	0.020339177	-0.018200463	<0.005
G11_1236_baseline_onlySCG	0.367292715	0.007727409	G11_1329_hypoxemia_onlySCG	0.405629809	0.01912868	-0.038337094	<0.005
Total mean	0.356543112	0.006779993		0.352709349	0.015983948	0.003833763	

Fig. 31. RTc results from baseline and hypoxemia including mean, standard deviation, difference and p-value.

Baseline			Hypercapnia			Comparison	
Pig	Mean	Std. dev	Pig	Mean	Std. dev	Difference	p-value
G2_0840_baseline_onlySCG	-0.045540541	0.036388388	G2_0857_hypercapnia_onlySCG	-0.043987097	0.067407463	-0.001553444	0.852914567
G2_1137_baseline_onlySCG	-0.002891892	0.014879026	G2_1139_hypercapnia_onlySCG	-0.029253165	0.011855788	0.026361273	<0.005
G2_1207_baseline_onlySCG	0.011027027	0.007312419	G2_1208_hypercapnia_onlySCG	-0.028819876	0.017183527	0.039846903	<0.005
G2_1243_baseline_onlySCG	0.01112	0.016966562	G2_1245_hypercapnia_onlySCG	-0.009015075	0.025293325	0.020135075	<0.005
G4_0923_baseline_onlySCG	-0.04384	0.029060272	G4_1003_hypercapnia_onlySCG	-0.03924	0.034869209	-0.0046	0.256732484
G5_0845_baseline_onlySCG	-0.053153846	0.017533188	G5_0902_hypercapnia_onlySCG	-0.116402299	0.032549459	0.063248453	<0.005
G5_0845_baseline_onlySCG	-0.053153846	0.017533188	G5_0928_hypercapnia_onlySCG	-0.1216125	0.027071298	0.068458654	<0.005
G7_0855_baseline_onlySCG	-0.049344	0.026945316	G7_1042_hypercapnia_onlySCG	-0.051992701	0.015407051	0.002648701	0.214741129
G8_1341_baseline_onlySCG	-0.031652968	0.036395737	G8_1413_hypercapnia_onlySCG	-0.042515942	0.031021755	0.010862974	<0.005
G9_0900_baseline_onlySCG	-0.036835443	0.007901666	G9_0923_hypercapnia_onlySCG	-0.040958904	0.032098405	0.004123461	0.02731425
G9_0900_baseline_onlySCG	-0.036835443	0.007901666	G9_1000_hypercapnia_onlySCG	-0.072378855	0.018801645	0.035543412	<0.005
G10_0849_baseline_onlySCG	-0.009714286	0.008588336	G10_0909_hypercapnia_onlySCG	-0.062614679	0.030854022	0.052900393	<0.005
G10_0849_baseline_onlySCG	-0.009714286	0.008588336	G10_0948_hypercapnia_onlySCG	-0.055264574	0.025066344	0.045550288	<0.005
G10_1031_baseline_onlySCG	-0.009724138	0.006882348	G10_1045_hypercapnia_onlySCG	-0.065144144	0.031370028	0.055420006	<0.005
G11_1236_baseline_onlySCG	-0.043687075	0.031455732	G11_1256_hypercapnia_onlySCG	-0.071928571	0.034698038	0.028241497	
G11_1236_baseline_onlySCG	-0.043687075	0.031455732	G11_1345_hypercapnia_onlySCG	-0.108044944	0.025335672	0.064357869	
Total mean	-0.027976738	0.019111744		-0.059948333	0.028805189	0.031971595	

Fig. 32. S2onset-Tend results from baseline and hypercapnia including mean, standard deviation, difference and p-value.

Baseline			Hypoxemia			Comparison	
Pig	Mean	Std. dev	Pig	Mean	Std. dev	Change	p-value
G2_1121_baseline_onlySCG	-0.014732984	0.029369639	G2_1125_hypoxemia_onlySCG	-0.027303371	0.032501162	0.012570386	<0.005
G2_1137_baseline_onlySCG	-0.002891892	0.014879026	G2_1153_hypoxemia_onlySCG	-0.004836364	0.015966539	0.001944472	0.37508856
G2_1225_baseline_onlySCG	0.028568182	0.044849633	G2_1229_hypoxemia_onlySCG	-0.017737705	0.019585177	0.046305887	<0.005
G4_0923_baseline_onlySCG	-0.04384	0.029060272	G4_0925_hypoxemia_onlySCG	-0.014137931	0.040836236	-0.029702069	<0.005
G5_0845_baseline_onlySCG	-0.053153846	0.017533188	G5_0915_hypoxemia_onlySCG	-0.054539216	0.033023106	0.00138537	0.613242141
G5_0949_baseline_SCG+PCG	-0.049145455	0.025125679	G5_0951_hypoxemia_SCG+PCG	-0.038313725	0.018851382	-0.010831729	<0.005
G6_0932_baseline_onlySCG	-0.041565891	0.013591283	G6_0939_hypoxemia_onlySCG	-0.026357895	0.019315216	-0.015207997	<0.005
G7_0855_baseline_onlySCG	-0.049344	0.026945316	G7_1025_hypoxemia_onlySCG	-0.020629213	0.018903091	-0.028714787	<0.005
G8_1341_baseline_onlySCG	-0.031652968	0.036395737	G8_1346_hypoxemia_onlySCG	-0.024168675	0.022095561	-0.007484293	0.019691687
G8_1518_baseline_onlySCG	-0.02405042	0.006925571	G8_1534_hypoxemia_SCG+PCG	-0.103866667	0.023401562	0.079816246	<0.005
G9_0900_baseline_onlySCG	-0.036835443	0.007901666	G9_0946_hypoxemia_onlySCG	-0.020033613	0.045977137	-0.01680183	<0.005
G9_1102_baseline_onlySCG	-0.02405042	0.006925571	G9_1104_hypoxemia_onlySCG	-0.103866667	0.023401562	0.079816246	<0.005
G10_0849_baseline_onlySCG	-0.009714286	0.008588336	G10_0855_hypoxemia_onlySCG	-0.027213333	0.037597584	0.017499048	<0.005
G10_0849_baseline_onlySCG	-0.009714286	0.008588336	G10_0936_hypoxemia_onlySCG	-0.038744186	0.038750863	0.0290299	<0.005
G10_1031_baseline_onlySCG	-0.009724138	0.006882348	G10_1033_hypoxemia_onlySCG	-0.019422819	0.018793753	0.009698681	<0.005
G11_1236_baseline_onlySCG	-0.043687075	0.031455732	G11_1241_hypoxemia_onlySCG	-0.065763889	0.046746151	0.022076814	<0.005
G11_1236_baseline_onlySCG	-0.043687075	0.031455732	G11_1329_hypoxemia_onlySCG	-0.125964602	0.026458839	0.082277527	<0.005
Total mean	-0.027013059	0.020380768		-0.043111757	0.028364995	0.016098698	

Fig. 33. S2onset-Tend results from baseline and hypoxemia including mean, standard deviation, difference and p-value.

APPENDIX IV COLLABORATION PORTFOLIO

This collaboration portfolio's purpose is to describe and reflect the work process on the article "Non- invasive diagnosis of pulmonary hypertension: The role of ECG and SCG in assessing S2 deviations". This analysis will give an insight into the working structure, time management, and reflection on the experiences gained in this project.

A. FSK- Professional social contract

The group made a FSK, which was a contract that includes a set of rules, to ensure conformity in the work done on the project, and to minimize conflicts that could arise. The main points were about project meetings, ambitions and what to expect of each other. To reduce conflicts and frustrations, a warning system was used. If a group member did not comply with the FSK on several occasions, as a last resort the member could be excluded from the group. The group had a basis for how they should work and if each group member complied there was no occasion for conflict.

B. Welfare meeting

A welfare meeting was held each week to discuss goals for the upcoming week and evaluate the work of the previous week. This includes evaluating if the effort was acceptable and if the goals were achieved. The time management schedule was discussed at this meeting, to ensure the schedule was up to date. The meeting was used as a way to update any private matters which could affect the work effort. Included in the meeting was planned absence, if any group members were otherwise occupied, this was noted. An example of the template used for these meetings is shown in table IX.

	Uge 37	Uge 38	Uge 39	Uge 40	Uge 41	Uge 42	Uge 43	Uge 44	Uge 45	Uge 46	Uge 47	Uge 48	Uge 49	Uge 50	Uge 51
Litteratursøgning															
Problemanalyse															
Problemafgrensning															
Problemformulering															
Hypoteser /krav?															
Design															
Statusseminar								01-nov							
Programmering															
Pilotforsøg															
Programmering															
Semcon													02-dec		
Analyse															
Statistik															
Artikel		Litteratursøgning		Hypoteser		Metoder		?		?		Abstract!			
Korrektur															
Aflever															

Fig. 34. Time management schedule Gantt

TABLE IX
WELFARE MEETING TEMPLATE

Item	Date: xx-xx-2024
Absent from the meeting:	
Project status:	
Last weeks project goals:	
Next weeks project goals:	
Evaluation of Gantt:	
Planned absence:	Monday: Tuesday: Wednesday: Thursday: Friday:
Next weeks plan:	Monday: Tuesday: Wednesday: Thursday: Friday:
Next weeks role (every 10. agenda):	Meeting recorder: Moderator: Daily agenda:
General well-being:	
Social events:	

C. Collaboration with supervisors

Throughout the project, a supervisor meeting was held approximately once a week. The meeting was student controlled, which means that the agenda was set by the students. This way, the students could pick discussion topics current for the project. Each meeting had a moderator and meeting recorder, to ensure that feedback and things discussed were notated.

D. Time management

The group used a gantt chart to keep track of milestones in the project work, as well as the time management to achieve the milestones, the gantt chart can be seen in figure 34. Different colors were used as indications: when work should be done (green), work in progress (yellow), and upcoming work (blue). The chart was altered regularly to fit the current situation of the project work. The weeks marked with red on the scheme were the last three weeks before the deadline of the project. The group could in these weeks extend their usual work time, to ensure that each member was satisfied with the outcome.

E. Agenda and work plan

An agenda and a work plan were made every day for the group to work from. These were made to accomplish an easier transition from one task to another and ensure productivity among the groupmembers. The work plan was a list of objectives yet to be made. Each group member could write their name beside the objective and mark it with a color, which indicated that the objective was a work in progress. After the objective was solved, it was marked green to show that it was done.

F. Sharing of knowledge

Throughout the project the group used peer learning and peer teaching, to ensure each member's knowledge of the subject was up to date. The sharing of knowledge could be done in different ways adaptive to the subject. Some method used were powerpoint presentations,

visualization of data, or explanation of the algorithm in plenary sessions. Throughout the project worksheets were made. These worksheets were also a method to peer learn. When a group member had been working on the algorithm, they wrote a comment about how it worked. This way the other group members could understand how the code was developed and functioned.

APPENDIX V SEARCH PROTOCOL

Project name: Non-Invasive Diagnosis of Pulmonary Hypertension: The Role of ECG and SCG in Assessing S2 Deviations

Persons involved and affiliations: Maibritt Post, Frederik Fruergaard, Natasja Garner, and Elisa Rytter

A. Definition of research subject

We would like to research how pulmonary hypertension leads to premature S2. This was accomplished by following the PICO-model:

PICO:

Patient group: Patients with cardiovascular diseases.

Intervention: ECG and SCG.

Comparison/Control: Compare healthy subjects and patients with pulmonary hypertension.

Outcome: An algorithm to detect premature S2 based on ECG and SCG.

B. Definition of inclusion and exclusion criteria

Inclusions and exclusions criteria are defined to narrow the literature search for things relevant to this research subject. Table X shows the inclusion and exclusion criteria which have been defined.

TABLE X
INCLUSION AND EXCLUSION CRITERIA

Inclusion criteria	Exclusion criteria
English and nordic language FullText Peer-reviews	Children <18 years Case studies Animal studies Surgical implants Disease / Syndrome Tumors

C. Search terms for subject aspects

In table XI the chosen aspects are shown, including the exclusions criteria, which have been defined.

TABLE XI
LIST OF ASPECTS

Aspect 1	Aspect 2	Aspect 3	Aspect 4
S2 OR "Second heart sounds" OR "Aortic valve" OR "Pulmonic valve" OR Semilunar	(SCG OR Seismocardiogram) OR (ECG OR Electrocardiogram) OR (PCG OR Phonocardiography)	"Pulmonary hypertension" OR "Pulmonary arterial hypertension"	NOT children NOT kids NOT adolescents NOT child NOT kid NOT pregnant NOT pregnancy

D. Selection of relevant sources

In table XII the selected sources are shown. The source selection also shows the reason as to why, the specific sources have been chosen.

TABLE XII
SOURCE SELECTION

Source	Reason for selection of source	Provider
PubMed	Contains medical and naturescience publications.	AUB
Embase	Contains biomedical and pharmalogical research.	AUB

E. PRISMA of search protocol

In figure 35 a Preferred Reporting Items for Systematic Reviews and Meta-analysis (PRISMA) of the search protocol is shown. This PRISMA shows the number of articles found in both search engines. The duplicates have been removed by using Rayyan. In the screening process, the articles were distributed between the group members to screen based on title and abstract. During the full text eligibility the articles were distributed between the members, and a summary for each article was written down. These summaries were looked through by the whole group to decide exclusion and inclusion of the articles.

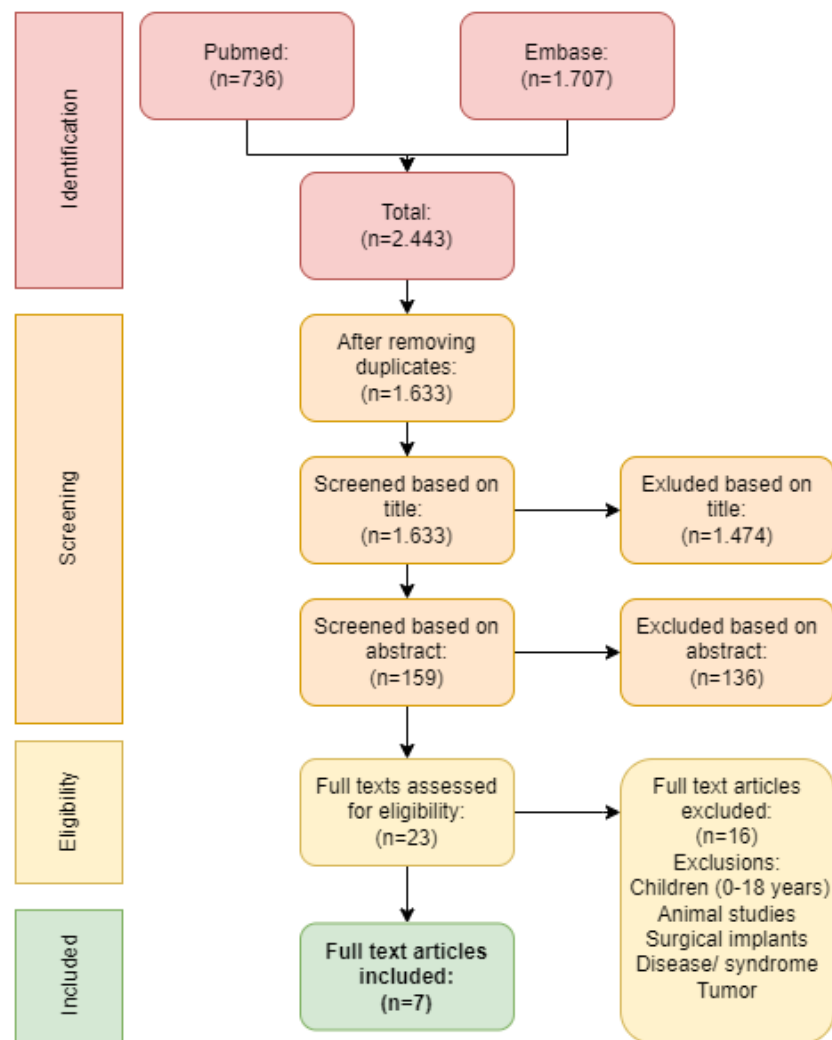


Fig. 35. PRISMA of the search protocol, with number of exclusions and inclusions.